

Species Distribution Modeling in the UK using presence-only data

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Abstract

Conserving biodiversity is a growing priority globally. The United Kingdom is no exception. It faces a continued decline in endangered species due to loss of habitat, urbanization, and climate change. This is where Species Distribution Modeling (SDM) comes in. It is an essential tool for predicting the suitability of habitats for these species. However, SDM faces challenges of its own. This is largely because species absence data is hard to obtain, making it difficult to prioritize conservation reserves, such as buying or protecting specific sites. Relying on presence-only data brings into light significant challenges, such as sampling bias and limited reliability of traditional modeling approaches. This dissertation aims to apply presence-only SDMs to model the distribution of habitat suitability of endangered species in the UK. The resulting habitat suitability maps serve as a decision-support tool, highlighting key areas to help guide conservation efforts like habitat restoration, monitoring, and potential reintroduction programs. Five species, according to the List of UK BAP Priority Terrestrial Mammal Species mentioned in the UK Biodiversity Action Plan, were selected: *Erinaceus europaeus* (Hedgehog), *Sciurus vulgaris* (Red Squirrel), *Muscardinus avellanarius* (Hazel Dormouse), *Lepus europaeus* (Brown Hare), *Pipistrellus pygmaeus* (Soprano Pipistrelle Bat).

The data was sourced through The Global Biodiversity Information Facility (GBIF) and relevant environmental variables were sourced from Google Earth Engine and The UK Centre for Ecology & Hydrology (UKCEH). After data cleaning, preprocessing and bias mitigation, multiple modeling approaches were applied, including MaxEnt, DeepMaxEnt, and ensembling. The bias mitigation was done through Target Group Background points. Model evaluation included using AUC-ROC, Boyce Index and weighted average between the two. This resulted in probability maps of suitable habitat. These maps revealed that: DeepMaxEnt ensembles consistently demonstrated superior performance compared to traditional MaxEnt, identifying Scotland and Northern Ireland as strongholds for the *Sciurus vulgaris*, while England and Wales provided primary suitability for the *Muscardinus avellanarius* and the *Erinaceus europaeus*. The *Lepus europaeus* shows fragmented suitability and the *Pipistrellus pygmaeus* showed a broad range of suitability across the UK. Furthermore, Land Surface Temperature was identified as the primary environmental feature across all species, with specific land cover types and spectral indices driving local suitability. This research contributes to ecological conservation and computational

literature by demonstrating the effectiveness of advanced presence-only data modeling techniques. The findings allow for valuable insights for conservation planning to guide endangered species protection in the UK.

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Chapter 1

INTRODUCTION

1.1 Background and Motivation

Conserving biodiversity has become a global priority, particularly because many species face the growing threat of habitat loss, climate change, and urban activity. In the United Kingdom, there are numerous species that are in continued decline and are listed as endangered. Due to this, Species Distribution Modeling (SDM) has come up as a valuable tool, providing an understanding of where species are probable to occur. This is valuable from a conservation perspective because it allows for efficient allocation of limited resources, helping to target field surveys and identify key habitats for protection. This dissertation focuses specifically on modeling the distribution of selected endangered species in the UK using presence-only data. Presence-only data refer to instances where the species has been observed, but there is no information about where it is absent, unlike presence-absence data which contain information about both the cases. With presence-absence data available, general statistical methods can be employed, such as those mentioned by Corsi, Duprè, and Boitani (1999) and Elith (2006). Even if absence data are recorded, it could be doubtful, as mentioned in Anderson, Lew, and Peterson (2003). This proposes a unique challenge that requires the use of innovative modeling techniques that can make predictions regarding the presence of a species using these incomplete data. The academic benefits of this dissertation will contribute to the literature on presence-only SDMs and explore how different modeling techniques can be used for this problem. Moreover, the environmental benefits are that using these species distribution maps, conservationists can get a better idea of where certain species are more likely to be found, directly supporting conservation planning.

1.2 Problem Statement

The task of a modeling method is to predict environmental suitability for the species as a function of the given environmental variables (Phillips et al., 2006). Species Distribution Modeling in the UK is dependent on understanding many environmental factors of where the species live; however, this is complicated by the nature of available data. Most real-world species data contain little to no absence data. This limits the use of traditional SDM methods and makes them difficult to be reliable. Sampling bias, where data collection is concentrated in easily accessible areas, and observer bias can further skew the data (Phillips et al., 2009). Therefore, it is a critical need to develop and assess modeling techniques that work with presence-only data. The ultimate goal is mapping endangered species to support conservation in the UK by delivering probability maps of suitable habitats that allows conservationists to explore these distribution maps and better understand species habitat needs, while also considering the nature of presence-only data. A central goal of this project is to bridge the gap between advanced computational modeling and practical conservation policy. In an era where evidence-based decision-making is important, it is not enough to simply produce complex models. It is necessary to translate these outputs into clear, trustworthy, and interpretable tools that decision-makers can use with confidence to allocate resources and protect vulnerable habitats.

Chapter 2

LITERATURE REVIEW

A review of existing literature on SDM using presence-only data reveals a range of methodologies employed worldwide to address the challenges of the nature of this data.

2.1 Common SDM algorithms for presence-only data

There are many algorithms that have been developed to address the challenge of SDM using presence-only data. Each of them has varying performance and underlying foundations. Max-Ent (Maximum Entropy) is a very popular method because of its ability to estimate the probability distribution of maximum entropy constrained by environmental variables, which shows better discrimination capacity compared to older methods like Genetic Algorithm for Rule-Set Prediction (GARP) (Phillips et al., 2006). When absence data are unavailable, Max-Ent is very effective in identifying suitable habitats and has been extensively used in SDM across different taxons and regions (Elith & Leathwick, 2009). Traditional models such as BIOCLIM and DOMAIN are computationally cheap but generally poor in performance. BIOCLIM works by creating a simple environmental envelope based on the minimum and maximum environmental values at known presence locations, while DOMAIN calculates a similarity metric between potential sites and known sites. However, BIOCLIM has the tendency to underpredict distribution ranges while DOMAIN often over-predicts (Sarquis et al., 2018). Recent advances have included ensemble and machine learning approaches such as Random Forest (RF) and XGBoost. In a comprehensive bench- mark study using 225 species, Valavi et al. (2022) found that RF and XGBoost, when properly tuned and corrected for bias, significantly outperformed classical methods like BIOCLIM and DOMAIN. While MaxEnt

performs reliably, recent innovations in deep learning have led to DeepMaxent. This is a neural network-based SDM that builds upon MaxEnt's foundational entropy principle. This model has shown to improve upon the traditional MaxEnt across all evaluated regions and biological groups, particularly outperforming other approaches under conditions of sampling bias (Ryckewaert et al., 2025).

2.2 Addressing Bias in Presence-Only Data

A major limitation of presence-only species distribution models (SDMs) is their vulnerability to sampling bias, which occurs when there is an uneven distribution of observations across geo- graphical locations due to factors such as observer preferences. This negatively influences model outputs, which can lead to the inaccurate estimation of habitat suitability (Phillips et al., 2009). To address this, Phillips et al. (2009) proposed the use of Target Group Background (TGB) methods, where background points are selected from presence data points of ecologically similar species. However, TGB must be used with caution. Though it might reduce the spatial bias, it can bring in a new ecological bias if the background species differs significantly from the target species in environmental preference (Støa et al., 2018). Støa et al. (2018) allowed modelers to detect and evaluate the effects of sampling bias using Frequency-of-Presence (FoP) curves. Botella et al. (2020) further explored this issue by formally modeling the observation process and demonstrating how Uniform Background (UB) and Target Group Occurrence Background (TGOB) methods both carry specific pitfalls. UB might overemphasize areas where lots of sampling has occurred, leading to niche estimates that are biased toward those well-sampled regions. TGOB could misestimate the niche if it combines data from species that don't share the same ecological preferences.

2.3 Ensemble Modeling

Ensemble modeling has emerged as a powerful strategy in SDM, offering a way to combine the different strengths of individual algorithms by merging their predictions. This approach mitigates the limitations in relying on a single model (Araújo and New, 2007). By aggregating predictions across different SDMs, researchers can improve model reliability and accuracy (Hy- sen et al., 2022). Harris et al. (2023) showed that ensemble SDMs provided better

predictive accuracy on unseen data points than any single constituent model. Meller et al. (2014) revealed that ensemble-based networks were more effective in capturing rare species distributions than single-model predictors.

2.4 Validation and Evaluation Metrics

Evaluating the accuracy of SDMs is important. However, commonly used metrics like Kappa and TSS have limitations, especially when applied to presence-only data. The Kappa statistic is highly sensitive to prevalence (the proportion of presences to absences). Allouche, Tsoar and Kadmon (2006) demonstrated that Kappa's values are minimized when prevalence is very low or very high, making it an unreliable measure. Although TSS has been widely regarded as robust, Leroy et al. (2018) demonstrate that it is susceptible to bias by species prevalence, giving inflated scores for rare species.

Chapter 3

RESEARCH QUESTIONS

1. Where are the most environmentally suitable areas for the endangered species selected in the UK?
2. What are the key environmental conditions that conservation efforts should prioritize for the selected endangered species in the UK?
3. Implicit question: To what extent can advanced modeling techniques mitigate the data limitations inherent in presence-only datasets where true absences are unavailable?

Chapter 4

Methodology

4.1 Research Design and Workflow

The design and workflow of this research entails species selection, data collection for species occurrences and environmental variables, data preprocessing, sampling bias mitigation, modeling, evaluation and validation, mapping and visualization.

4.2 Species Selection and Data Sources

4.2.1 Species Selection

Initially, species from the The International Union for Conservation of Nature (IUCN) Red List in the UK were chosen as candidates. However, data relating to their UK presence was very limited. Candidate species were then chosen from the List of UK BAP Priority Terrestrial Mammal Species mentioned in the UK Biodiversity Action Plan (UK BAP Priority Species — JNCC - Adviser to Government on Nature Conservation, 2024). Chosen species: *Erinaceus europaeus* (Hedgehog), *Sciurus vulgaris* (Red Squirrel), *Muscardinus avellanarius* (Hazel Dormouse), *Lepus europaeus* (Brown Hare), *Pipistrellus pygmaeus* (Soprano Pipistrelle Bat). Natural England also provides detailed summaries of the drivers of change for each of these five species (A Review of the Population and Conservation Status of British Mammals (JP025), 2018).

4.3 Data Sources

Different data sources were used for getting the dataset for species occurrences and environmental variables for locations of these occurrences. The selection of species for Target Group Background analysis (see section 4.5) was based on expert opinion. Table 4.1 shows the different kinds of environmental factors considered that contribute towards species distribution.

Environmental Factor	Positive or Negative Influence	Details
Prey distribution	+ve	Source of food
Predator distribution	-ve	Consumer of species
Spectral Indices	variable	Differs due to species preferences
Temperature	variable	Differs due to species preferences
Land cover	variable	Differs due to species preferences
Road proximity	-ve	Possible threats for select species
Wind turbine proximity	-ve	Possible threat for flying species
Distribution of similar species	-ve	Evidence of observation which did not identify target species

Table 4.1: Summary table for environmental factors

4.3.1 Presence data

The Global Biodiversity Information Facility (GBIF)¹ was used to attain presence data for the species. GBIF is an international network and data infrastructure that provides open access to biodiversity data, aggregating records for numerous institutions, research projects, and citizen science initiatives which vary in their spatial coverage and collection methodologies. To ensure the temporal relevance of the data and avoid potential biases in recording effort caused by the COVID-19 pandemic lockdowns, the dataset was filtered to include only records from 2022

¹<https://www.gbif.org/>

onwards within the UK. This time-frame provided a sufficient number of occurrence records for robust modeling. The location coordinates and time of occurrences were then extracted.

***Erinaceus europaeus* - European hedgehog**

The data for *Erinaceus europaeus* was acquired from GBIF.org (24 May 2025a). *Erinaceus europaeus* have a predatory species, the Badger (*Meles meles*), which has inverse relationship to their occurrence (Parrott et al. (2014) and Trewby et al. (2014)). Their data was acquired from GBIF.org (24 May 2025b). Ground beetles are a common species upon which the European hedgehog prey upon according to Mathews et al. (2018). According to Baldwin (2022), beetles, especially ground beetles (carabids), are commonly found among hedgehog stomach contents. Species such as *Pterostichus madidus* (Black Clock Beetle), *Harpalus* (Genus of Ground Beetle), *Nebria* (Genus of Ground Beetle), *Amara* (Genus of Carabid Beetle), and Bembidion (Genus of Carabidae Beetle) have also been identified as part of their diet. Their data was sourced from GBIF.org (24 May 2025c) Road casualties are also likely to have had an important effect on local populations (Huijser and Bergers (2000) & Wembridge et al. (2016)). For getting road data, OSMnx is a Python library that simplifies the process of downloading and analyzing street networks from OpenStreetMap. It can be used to find the nearest road to a given point. *Talpa europaea* (European Mole), *Sorex araneus* (Common Shrew), and *Apodemus sylvaticus* (Wood Mouse) were chosen for Target Group Background points (see section 4.5) as these were the species with similar requirements. Their data was acquired from GBIF.org (15 June 2025a).

Environmental Factor	Influence	Details
Predator distribution (<i>Meles meles</i> (Badger))	-ve	Inverse relationship with <i>Erinaceus europaeus</i> occurrence.
Prey distribution (Ground beetles, e.g. <i>Pterostichus madidus</i> , <i>Harpalus</i> , <i>Nebria</i> , <i>Amara</i> , <i>Bembidion</i>)	+ve	Source of food.
Road proximity	-ve	Road casualties negatively impact local hedgehog populations.
Target Group Background species	-ve	<i>Talpa europaea</i> (European Mole), <i>Sorex araneus</i> (Common Shrew), and <i>Apodemus sylvaticus</i> (Wood Mouse) chosen as background species for modelling.

Table 4.2: Summary of environmental factors and species interactions influencing *Erinaceus europaeus* distribution.

***Sciurus vulgaris* - Red squirrel**

The data for *Sciurus vulgaris* was acquired from GBIF.org (24 May 2025d). The presence of Sitka spruce is unfavourable for red squirrels (Shuttleworth et al. (2012)). I got their presence data from GBIF.org (24 May 2025e). They also seem to be in competition with Grey squirrels (Gurnell et al. (2004) & Gurnell et al. (2015)). Their data was sourced from GBIF.org (24 May 2025f). *Tamias sibiricus*, *Glis glis*, and *Muscardinus avellanarius* were chosen for Target Group Background points. Their data was acquired from GBIF.org (15 June 2025b).

Environmental Factor	Influence	Details
Presence of Sitka spruce	-ve	Unfavourable habitat.
Competition with Grey squirrels	-ve	Grey squirrels compete with red squirrels for resources.
Target Group Background species	-ve	<i>Tamias sibiricus</i> (Siberian Chipmunk), <i>Glis glis</i> (Edible Dormouse), and <i>Muscardinus avellanarius</i> (Hazel Dormouse).

Table 4.3: Summary of environmental factors and species interactions influencing *Sciurus vulgaris* distribution.

***Muscardinus avellanarius* - Hazel Dormouse**

The data for *Muscardinus avellanarius* was acquired from GBIF.org (24 May 2025g). Sarah et al. (2023) shows that the presence of hazel plants, Beech, Birk, Honeysuckle, Oak, and Hawthorn to be favorable to the Hazel Dormouse as it uses them in their nests. Hazel, Beech, Birk, Honeysuckle, Oak, and Hawthorn data was acquired from GBIF.org (24 May 2025h), GBIF.org (24 May 2025i), GBIF.org (24 May 2025j), GBIF.org (24 May 2025k), GBIF.org (24 May 2025l) and GBIF.org (24 May 2025m), respectively. *Eliomys quercinus*, *Apodemus sylvaticus*, and *Apodemus flavicollis* were chosen for Target Group Background points. Their data was acquired from GBIF.org (15 June 2025c) and GBIF.org (15 June 2025d).

Environmental Factor	Influence	Details
Presence of Hazel, Beech, Birk, Honeysuckle, Oak, and Hawthorn	+ve	These plant species provide food and nesting material.
Target Group Background species	-ve	<i>Eliomys quercinus</i> (Garden Dormouse), <i>Apodemus sylvaticus</i> (Wood Mouse), and <i>Apodemus flavicollis</i> (Yellow-necked Mouse).

Table 4.4: Summary of environmental factors and species interactions influencing *Muscardinus avellanarius* distribution.

***Lepus europaeus* - Brown hare**

The data for *Lepus europaeus* was acquired from GBIF.org (24 May 2025n). Changes to agricultural practices, particularly the switch towards winter wheat, may reduce food and shelter opportunities, Hewson and Hinge (1990). I got the data for winter wheat from GBIF.org (24 May 2025o). *Lepus timidus* and *Oryctolagus cuniculus* were chosen for Target Group Background points. Their data was acquired from GBIF.org (15 June 2025e).

Environmental Factor	Influence	Details
Winter wheat	-ve	Switch towards winter wheat may reduce food and shelter opportunities.
Target Group Background species	-ve	<i>Lepus timidus</i> (Mountain Hare) and <i>Oryctolagus cuniculus</i> (European Rabbit).

Table 4.5: Summary of environmental factors and species interactions influencing *Lepus europaeus* distribution.

***Pipistrellus pygmaeus* - Soprano Pipistrelle Bat**

The data for *Pipistrellus pygmaeus* was acquired from GBIF.org (05 August 2025a). According to their diets, they make use of the families Ceratopogonidae and Chironomidae (Barlow, 1997). The data for Ceratopogonidae and Chironomidae was acquired from GBIF.org (05 August 2025b) and GBIF.org (05 August 2025c), respectively. These bats are one of the primary species killed at wind turbines according to Mathews et al. (2016). The data for operational wind turbines was acquired from Boswarva (2023). The bat is also one of the primary species recorded in vehicle collisions according to Fensome and Mathews (2016). Road data was acquired from OpenStreetMap as was done for *Erinaceus europaeus*. The bat is also one of the species most frequently injured and killed by cats according to Mathews et al. (2018), specifically the *Felis Catus* (Khayat et al., 2020). The data for *Felis Catus* was acquired from GBIF.org (05 August 2025d). *Pipistrellus pipistrellus* and *Pipistrellus nathusii* were chosen for Target Group Background points. Their data was acquired GBIF.org (05 August 2025e).

Environmental Factor	Influence	Details
Prey availability (Ceratopogonidae, Chironomidae)	+ve	These insect families provide food.
Wind turbine proximity	-ve	Wind turbines cause mortality in bats.
Road proximity	-ve	Vehicle collisions are a significant threat.
<i>Felis catus</i> (Cat) presence	-ve	Domestic cats frequently injure or kill these bats.
Target Group Background species	-ve	<i>Pipistrellus pipistrellus</i> (Common Pipistrelle) and <i>Pipistrellus nathusii</i> (Nathusius' Pipistrelle).

Table 4.6: Summary of environmental factors and species interactions influencing *Pipistrellus pygmaeus* distribution.

4.3.2 Land Cover

Land cover is defined as the observed physical cover on the earth's surface, including vegetation (natural or planted) and human constructions. Particular types of land cover (e.g. grassland or marsh) are preferred by some species as it contributes to their feeding and/or nesting. It was sourced from The UKCEH Land Cover Map for 2023 for Northern Ireland (NERC EDS Environmental Information Data Centre, 2024) and The UKCEH Land Cover Map for 2023 for Great Britain (NERC EDS Environmental Information Data Centre, 2024a). It is the eleventh UK land cover map produced by UKCEH Land cover is given as 21 UKCEH Land Cover Classes based on Biodiversity Broad Habitats (Jackson, 2000).

4.3.3 Spectral Indices

Spectral indices are numerical values calculated from the spectral bands of an image to highlight specific features, such as vegetation health, soil moisture, or water content. These variables can help to distinguish the niche suitable for different species. This was sourced from COPERNICUS/S2_HARMONIZED (Harmonized Sentinel-2 MSI: MultiSpectral Instrument, Level-1C (TOA), 2025).

4.3.4 Land Surface Temperature

Land surface temperature (LST) is defined as a measure of how hot or cold the surface of the Earth would feel to the touch (Guillevic et al., 2018). Most species have a preferred temperature range for nesting in particular, or are associated with food plants or prey which only occur at certain temperature ranges. It was sourced from MODIS/061/MOD11A1 (MOD11A1.061 Terra Land Surface Temperature and Emissivity Daily Global 1km, 2025).

4.4 Data Cleaning, Feature Engineering, and Preprocessing

Data cleaning involved removal of duplicate rows, removal of null values and data type standardization. A number of features were engineered. All features that involved presence of predator and prey species were engineered on a proximity basis. For example, the proximity feature for Badgers was engineered by checking if the location coordinate for a Badger was within 1km of the location coordinate of the presence of *Erinaceus europaeus* in the same month and year. This was done using a Ball Tree. A Ball Tree is a data structure that is used for efficient nearest-neighbour search in a multidimensional space. It has the advantage of being extremely fast when dealing with high-dimensional datasets, such as the dataset used in this project. It is used when there exists a dataset of points and you want to find the nearest neighbours of a query point. This is where the Ball Tree comes in. It organizes the points to allow for faster searching, in logarithmic time; $O(\log n)$. This allowed for an efficient way to check if one species was in proximity of another during a given date and location. Then it was a simple matter of assigning a Boolean column of `badger_presence` where a value of 1 was given if it was within a proximity of 1km and 0 otherwise. The metric for distance used here was the Haversine metric which is the angular distance between two points on the surface of a sphere. The same procedure was applied to the following presence data: Carabids, *Pterostichus madidus*, *Harpalus*, *Nebria*, *Amara*, and *Bembidion* for *Erinaceus europaeus*; Sitka spruce and Grey squirrel for *Sciurus vulgaris*; Hazel, Beech, Birch, Honeysuckle, Oak, and Hawthorn for *Muscardinus avellanarius*; Winter wheat for *Lepus europaeus*; and Ceratopogonidae, Chironomidae, *Felis catus*, and Wind turbines for *Pipistrellus pygmaeus*. For road data, a script was written using OpenStreetMap data. this script calculated the distance from each location coordinate in the dataset to the nearest road. For this, only non-pedestrian roads were chosen. It first divides the UK land area into small tiles of 1km x

1km. Then, a road network was downloaded for each tile and both the roads and the location coordinates were projected into a metric coordinate system, computing the Euclidean distance from each coordinate to its nearest road segment. Any coordinates without a nearby road are assigned null values. Then the road distance is transformed into a Boolean column called `near_road` where if the distance is less than or equal to 500 meters, it is assigned a value of 1, and 0 otherwise. This was used for *Erinaceus europaeus* and *Pipistrellus pygmaeus*. For land cover data, it was determined to be the mode of the land covers within a 1km x 1km area of a location coordinate and within one month of a species's detection. For each spectral index, the value was determined to be the mean of the specific indices within a 1km x 1km area of a location coordinate and within one month of a species's detection. For the land surface temperature, it was determined to be the mean of the land surface temperatures within a 1km x 1km area of a location coordinate and within one month of a species's detection.

To preprocess the data for modeling, the Boolean data columns were transformed to the Boolean data type. The Land cover feature was one-hot encoded. One-hot encoding refers to converting categorical data types into binary types. It is done for nominal data, where the categories do not have an inherent order. For example, colors are of a nominal data type, as red, black, yellow, purple, etc. do not have an ordering. Label encoding, such as assigning 0 to red, 1 to black, 2 to yellow and 3 to purple, is not recommended, as machine learning models tend to learn the the relationships between the distances of these values. Hence, it will learn that black and yellow are similar due to having a distance of 1 (2-1), and that red and purple are very different due to having a distance of 3 (3-0). In reality, these are bogus relationships. In contrast, sizes such as small, medium, and large have an ordering and are referred to as an ordinal data type. Ordinal data types can undergo label encoding as the distance between their elements makes sense and the model can learn valuable relationships from them. For nominal data types, we use one-hot encoding, where a column containing red, black, yellow, and purple would be split into four columns, each being a Boolean binary column having either a value of 0 or 1. This way, each record in the data would have either [0,0,0,1], [0,0,1,0], [0,1,0,0], or [1,0,0,0] for the color column, indicating if this color is attributed to that record or not, for all the colors. Similarly, one-hot encoding is applied to the land cover data because of its nominal nature. The same processes were applied to the Target Group Background species.

4.5 Bias Handling

Survey efforts for recording presences are uneven. Some areas might be visited frequently, and others rarely. This can introduce bias into our machine learning model. SDMs require both presence and background points. Background points refer to pseudo-absences, where the species may or not be present. They are required as true absences do not exist. Observers usually record only what they see, not what they don't see. If you see a species somewhere, you can record it as present. But, what if you do not see it the next day? Would you classify it as an absence? How would you truly know if the species was supposed to be there or not? Would you call it an absence just because you saw it the previous day and now you do not? This introduces bias. Even if we take all locations without a presence record and call them 'absences', we bring in false absences which can also introduce bias. To address this, the use of Target Group Background (TGB) points is employed, where background points are selected from presence data points of ecologically similar species. Instead of sampling random background points from all over the landscape, we sample from locations where presence records for ecologically similar species are available. The key idea is that these locations were surveyed and a species was detected, but the target species was not observed. This represents a more informed concept of absence; it reflects areas where sampling actually occurred, rather than assuming that unvisited locations without records are true absences.

Figures 4.1 shows the contrast between the presence points and background points of *Erinaceus europaeus*, *Sciurus vulgaris*, *Muscardinus avellanarius*, *Lepus europaeus*, and *Pipistrellus pygmaeus*. The red points are presence points and the blue points are background points.

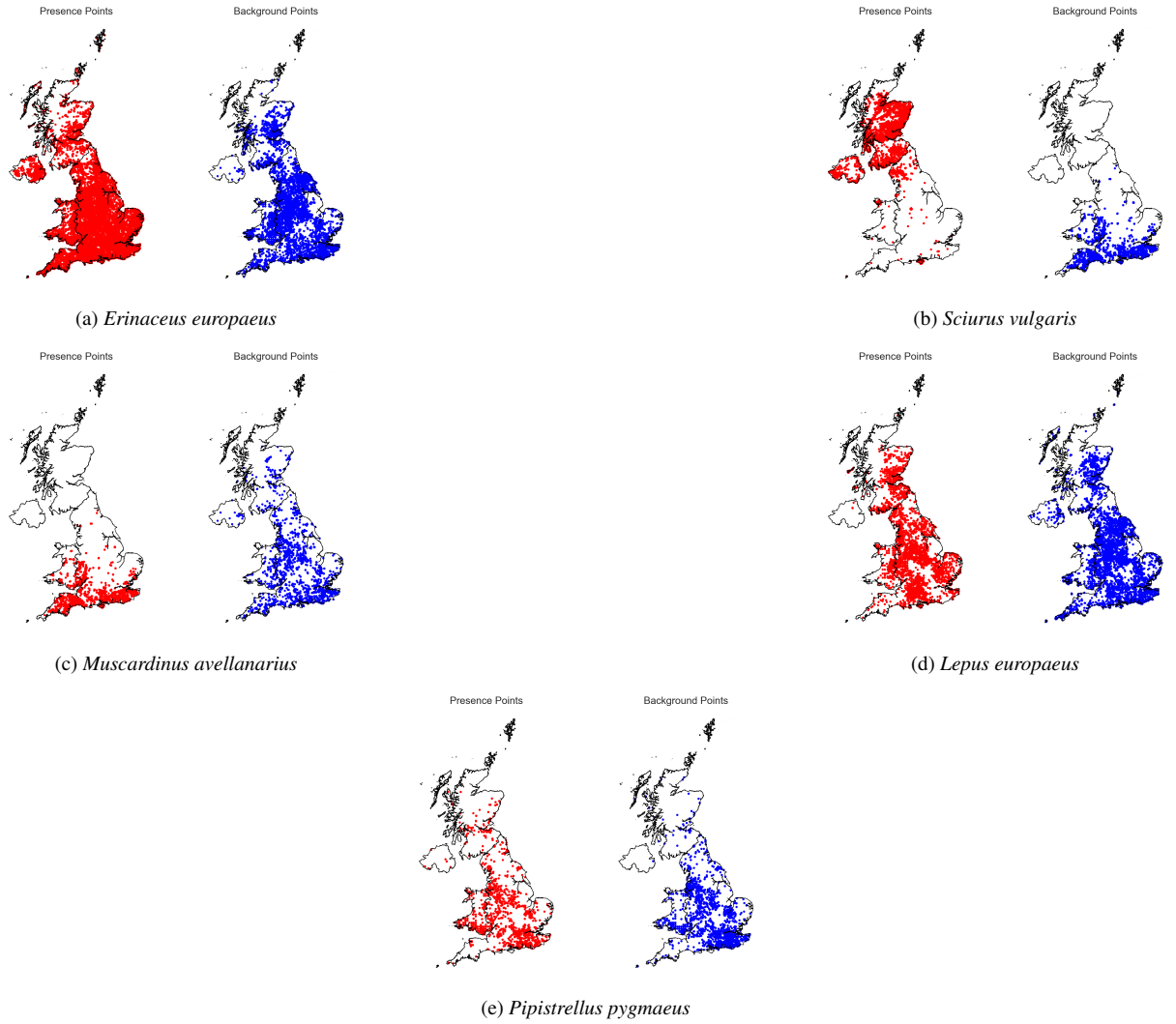


Figure 4.1: Presence vs background for multiple species

4.6 Modeling

As said previously, SDMs require presence data and background data. These contrasting pairs allow an SDM to learn the differences in relationships between locations that are more suitable to a species and locations that are less suitable, according to their environmental factors. All the modeling methods used a training set of 80% of the dataset and 20% of the validation set. The numerical features were then scaled using a standard scaler according to the formula given in Section A.1. After this transformation, all numerical features will have a mean of 0 and a standard deviation of 1. This is needed for machine learning as raw features can have different scales. This standardization helps in regularization, optimization and provides equal importance of the features since many models are sensitive to scale. Without scaling, a feature with a large numeric range will dominate, in importance, a feature of lower numeric range. Hence, standardization puts all features on the same ground.

4.6.1 Maxent

Introduction to Maxent

According to Phillips et al. (2006), MaxEnt is a popular method for Species Distribution Modeling because of its ability to estimate the probability distribution of maximum entropy constrained by environmental variables. Maxent is a general-purpose method for making predictions or inferences from incomplete information. Its origins lie in statistical mechanics (Jaynes, 1957). "The idea of Maxent is to estimate a target probability distribution by finding the probability distribution of maximum entropy (i.e., that is most spread out, or closest to uniform), subject to a set of constraints that represent our incomplete information about the target distribution." (Phillips et al., 2006). Applying MaxEnt to presence-only data for species distribution modeling allows for the location coordinates of the study area (the UK in this case) to make up the space on which the probability distribution is defined. These coordinates comprise of different features which are fed in to the model, as defined in Section 4.4.

According to Phillips et al. (2006), MaxEnt has numerous advantages: (1) Only presence data is required (and its constituent environmental features) for the study area. (2) MaxEnt is able to understand both numerical features and categorical features, and understand the underlying relationships between them. (3) The MaxEnt probability distribution function is able to be defined using precise mathematical notation, which allows for quantitative and qualitative analysis. (4) The output probability is continuous, which allows for subtle differences in habitat suitability to be captured across areas. The unknown probability distribution, formally denoted as π , is over a finite set X , which is the set of location coordinates in the study area. These individual coordinates of X are referred to as points. The distribution π assigns a non-negative probability $\pi(x)$ to each point x , with all the probabilities summing up to 1. π is approximated with another probability distribution, which is denoted as $\hat{\pi}$. The entropy of $\hat{\pi}$ is

defined as:

$$H(\hat{\pi}) = - \sum_{x \in X} \hat{\pi}(x) \ln \hat{\pi}(x)$$

where $\ln$ denotes the natural logarithm. The minus sign ensures the output entropy to be positive and its maximum value would be the natural logarithm of the number of elements in X .

Implementation

”The crucial point is that, suitably construed, logistic regression is the same thing as maxent with just two candidates” (Hayes, 2012). ”The penalised logistic regression outperformed standard logistic regression and equalled the performance of Maxent” (Gastón et al., 2011). Moreover, Berger et al. (1996) shows that the maximum entropy distribution subject to feature-based constraints is directly equivalent to the form of exponential functions, and that finding this distribution corresponds to the equation of maximizing the likelihood (equation given in Section A.9). This equation is exactly the form of multinomial logistic regression. Hence, logistic regression can be used for finding maximum entropy. The prediction probability is what is used as the probability of habitat suitability given the environmental factors and other features such as prey and predator presence, road proximity, etc, as defined in Section 4.4. The presence points are treated as the positive class and the background points are treated as the negative class.

4.6.2 DeepMaxEnt

While MaxEnt performs reliably, recent innovations in deep learning have led to DeepMaxEnt. This is a neural network-based SDM that builds upon MaxEnt’s foundational entropy principle. This model has shown to improve upon the traditional MaxEnt across all evaluated regions and biological groups, particularly outperforming other approaches under conditions of sampling bias (Ryckewaert et al., 2025). DeepMaxEnt uses neural networks to learn the relationships between features for the species. It models presence probabilities across different locations with a neural network using a normalized Poisson loss (equation found in Section A.3). Deep learning allows for learning feature representations automatically without the need to explicitly define them, saving efforts. It learns complex, non-linear relationships between environmental features and species presence. With help from Mr. Maxime Ryckewaert, two different losses were tested with DeepMaxEnt on the dataset. A loss function is used for binary

classification and computes the loss that measures how well the predictions match the true labels. The goal is to minimize this loss.

Binary Cross-entropy with Logits Loss

The Binary Cross-entropy with Logits Loss (BCEWithLogitsLoss) is the binary cross-entropy loss applied directly to logits (pre-sigmoid scores, which are the raw outputs of the model before applying the sigmoid function). The equation for this can be found in Section A.4.

DeepMaxEnt Loss

This loss implements binary cross-entropy loss with probabilities computed from logits via a sigmoid, adding a small ϵ for numerical stability. The equation for this can be found in Section A.5.

4.6.3 Ensemble Modeling

Ensemble modeling is an approach used in machine learning where the results of multiple machine learning models are combined to make a single prediction. The main idea is that a group of models usually have a better performance than that of a single one, as individual models may have different strengths and weaknesses. This has many advantages: (1) It reduces overfitting as it introduces variance on a model-level. (2) It improves predictive accuracy. (3) It makes models more robust to noise in the data. There are many ways to build ensembles of models. Usually, the way to do it is using voting-based ensembles.

Hard Voting

Hard voting means that each model in the ensemble predicts a class label (e.g. 0 or 1). The final prediction is the class that has the majority of votes. For example, there are three classifiers for predicting a sample. Model A predicts 1; model B predicts 0; and model C predicts 1. Using hard voting, the majority is 1 and hence, the final prediction is 1. This method does not need probability outputs from individual models; this causes the ignorance of the confidences of the individual models.

Soft Voting

Soft voting means that each model outputs class probabilities rather than the final labels. The ensemble averages these individual probabilities (usually using mean, median or weighted

averaging), and the final class is the one with the highest average probability. For example, there are three classifiers for predicting a sample. Model A predicts a probability of 0.8 for class 1; Model B predicts a probability of 0.3 for class 1; Model C predicts a probability of 0.9 for class 1. The average probability (using mean) is $\frac{0.8+0.3+0.9}{3} = 0.67$. Using soft voting, the prediction is 1, since $0.67 > 0.5$ (assuming the threshold of class division is 0.5). The formal equation of soft voting can be found in Section A.6. Since the final outputs in this project are the predicted probabilities of suitable habitats instead of hard labels, soft voting was used.

4.6.4 Implementation

Validation metrics

Since the dataset consists of presence and background data, true absences do not exist. Therefore, using common metrics like precision, recall, or F1 can be misleading. These metrics assume the background points are true negatives, which we can't guarantee in presence-only modeling. That is why most MaxEnt studies use AUC-ROC, or metrics like the Boyce Index for independent validation.

AUC-ROC (Area Under the Receiving Operating Characteristic)

AUC-ROC measures discrimination: how well the model ranks presence points above background points. It defines the True Positive Rate (TPR, also known as *sensitivity*) and False Positive Rate (FPR, also known as $1 - \textit{specificity}$) across all thresholds. The equations of *sensitivity* and *specificity* can be found in Section A.7. The final ROC curve is TPR vs FPR; the AUC is the area under the curve. AUC corresponds to the probability that a randomly chosen presence point has a higher predicted score than a randomly chosen background point. The range of AUC goes from 0 to 1. This means that the more the area under the curve (the higher the value), the better the model is able to separate the classes, independent of the threshold.

Boyce Index

Boyce Index asks the question: do presence points occur more often than expected in high-suitability regions? It is computed as follows: (1) Model suitability scores $s(x)$ are calculated for presence points and background points. (2) The suitability range is partitioned into bins. (3) For each bin, the P value is calculated, which is the fraction of presence points in that bin. (4) For each bin, the E value is calculated, which is the fraction of background points

in that bin. (5) For each bin, $\frac{P}{E}$ is calculated. (6) Then, the relationship between the suitability bin midpoints and the $\frac{P}{E}$ for each bin is summarized, using Pearson correlation coefficient (equation found in Section A.8). The range of the Boyce Index ranges from -1 to +1, where higher positive values indicate better performance, 0 indicates random performance, and negative values indicate an inverse (worse than random) relationship.

Approach

For each species, two separate MaxEnt models were trained and tuned for their hyperparameters: (1) Pipeline A (AUC-ROC tuned): hyperparameters were picked that maximised validation AUC-ROC. (2) Pipeline B (Boyce tuned): hyperparameters were picked that maximised validation Boyce Index. This way ended up in having two 'best' models per species, each optimized for a different objective. Then cross-metric auditing was employed. The following was done for each species: for the Boyce tuned best model, both Boyce Index and AUC-ROC were measured on the same validation set. The same was done for the AUC-ROC tuned best model. Then the metrics were combined by averaging to get a single joint score per model. This was done by first normalizing the AUC and BI values to a comparable scale, then their average was calculated. This new merged metric provides a single, unified measure of the model performance for each species by capturing both the discrimination performance and calibration performance of the model. A high value of this merged metric indicates that the model not only correctly ranks presence points relative to background points but also accurately identifies high-suitability areas where the species occur. It gives higher scores to locations where the species is actually present compared to random background points, and it predicts the areas where the species is most likely to occur with higher suitability scores. This metric provides a balanced view allowing comparisons across models by integrating two complementary metrics of predictive performance into a single score that is interpretable. Graphs were then plotted for these two models to compare their performance (found in Section B.2). Additionally, a third model was built by ensembling the two tuned models; using soft-voting (probability averaging with equal weightage) since the outputs of both the models were continuous probabilities. A graph was then plotted for this ensemble model as well to compare its performance with the single models (found in Section B.2). This soft-voted ensemble often dominated either (or both) of the single models on average across species, as expected.

These performances from the models were good, but they could have been improved by a noticeable margin if DeepMaxEnt were to be used, given the capabilities of deep learning. This DeepMaxEnt network was built using the following architecture: the inputs were the features defined in Section 4.4. The backbone of the network consists of two hidden layers, with residual connections. The output layer is one neuron (logit z), the probability given as $p = \sigma(z)$. This architecture was taken from Ryckewaert et al. (2025), and through Mr. Maxime Ryckewaert himself. For regularization, early stopping was used and Adam was used as the optimizer with L2 regularization to combat overfitting. Model training was done for each species using the two loss functions separately and each of them was optimized under different evaluation metrics: namely AUC and Boyce Index. This produced four models for each species: (1) a model trained using BCELogitsLoss optimized under Boyce Index. (2) a model trained using BCELogitsLoss optimized under AUC. (3) a model trained using DeepMaxEnt Loss optimized under Boyce Index. (4) a model trained using DeepMaxEnt Loss optimized under AUC. After these individual models were trained, experiments were done using ensembling modeling. A power set over the four trained models was defined per species for each possible combination of the models (from one model to all four models together). The predicted probabilities were averaged across each possible combination. The AUC and Boyce Index were calculated for each combination, with their averages (after normalizing them) to produce a mixed score. These combinations were then ranked according to the mixed score and visualized using a grouped bar plot for each species, showing the values of these three metrics across the ensemble combinations. These figures can be found in Section B.3. As expected, the DeepMaxEnt models outperformed the MaxEnt models. Table 4.7 shows the results of the DeepMaxEnt models picked for each species, with their mixed scores. However, there is one thing to note: the MaxEnt models consistently gave a better mixed score for the *Pipistrellus pygmaeus* (Soprano Pipistrelle Bat) but the AUC was severely low for all of them. The mixed score was only inflated because of the high Boyce Index. Hence, the DeepMaxEnt was chosen for *Pipistrellus pygmaeus* (Soprano Pipistrelle Bat) which has an overall lower mixed score, but the AUC and Boyce Index are relatively balanced.

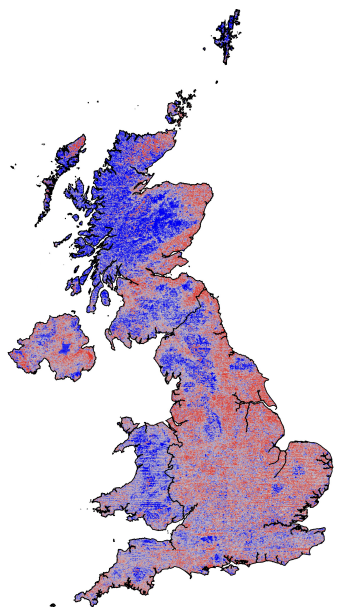
Species	Model(s)	Score
<i>Erinaceus europaeus</i> (Hedgehog)	BCEWithLogitsLoss tuned on Boyce Index + DeepMaxEnt Loss tuned on Boyce Index	0.810
<i>Sciurus vulgaris</i> (Red Squirrel)	DeepMaxEnt Loss tuned on Boyce Index + DeepMaxEnt Loss tuned on AUC + BCEWithLogitsLoss tuned on AUC + BCEWithLogitsLoss tuned on Boyce Index	0.798
<i>Muscardinus avellanarius</i> (Hazel Dormouse)	DeepMaxEnt Loss tuned on Boyce Index + DeepMaxEnt Loss tuned on AUC + BCEWithLogitsLoss tuned on AUC + BCEWithLogitsLoss tuned on Boyce Index	0.756
<i>Lepus europaeus</i> (Brown Hare)	BCEWithLogitsLoss tuned on Boyce Index + BCEWithLogitsLoss tuned on AUC + DeepMaxEnt Loss tuned on AUC	0.747
<i>Pipistrellus pygmaeus</i> (Soprano Pipistrelle Bat)	BCEWithLogitsLoss tuned on AUC	0.378

Table 4.7: Model loss functions and corresponding scores for different species.

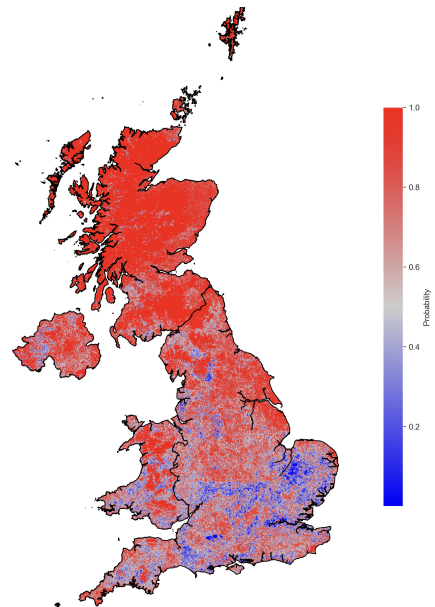
4.7 Visualization, Mapping Strategy and Results

A script was created with the primary goal of creating a high-resolution grid of points across the UK and save their geographic coordinates as the inference set. First, a detailed map of the UK was loaded from Office for National Statistics (2024). Then, the shapes of England, Scotland, Wales, and Northern Ireland were merged into a single, unified polygon. Then, a large, rectangular grid of points were created that completely covers the polygon of the UK. These points were of size 1km x 1km as all the environmental features were collected on a similar scale. Finally, filtering was applied on this grid to keep only the point that fall within the boundaries of the UK (to remove points located on external seas and oceans). Then, these points were converted to the standard latitude and longitude system and saved as the inference

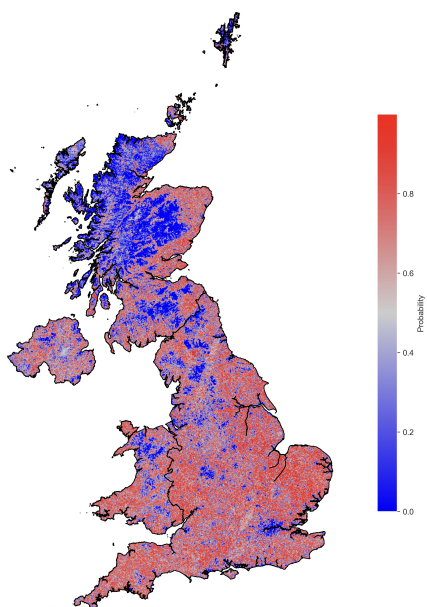
set. For each species, inference on these points was carried out. For the inference points, the same environmental features were collected, preprocessed and engineered as explained in Section 4.4 as of 1 May 2025. Hence, the output maps produced will be for this date. This date was selected due to lack of availability of data points past this cut-off. After this, the best models that were selected according to Table 4.7 were loaded and run on the inference data points, building an ensemble probability for each inference point where ensemble was applicable and building individual probability for each inference point where ensemble was not applicable. After gathering the probabilities for each inference data point, they were projected on to a spatial raster over the UK and plotted. This resulted in a species distribution probability map (also called habitat suitability map) for each of the species across the UK. The map was drawn with a color gradient from blue (low probability) to red (high probability), overlaid with UK country borders for reference to each of the four countries in the UK. Figure 4.2 shows the final habitat suitability probability maps for each of the five species. These maps are the final product of this project. The maps are a decision-support tool that highlight the conditions and areas which are most suitable to the endangered species, helping guide ecologists and conservationists to areas where they might be able to focus protection efforts, such as nature reserves, woodland management, monitoring, etc. to safeguard the remaining strongholds of these endangered species (more on this in Section 5.6).



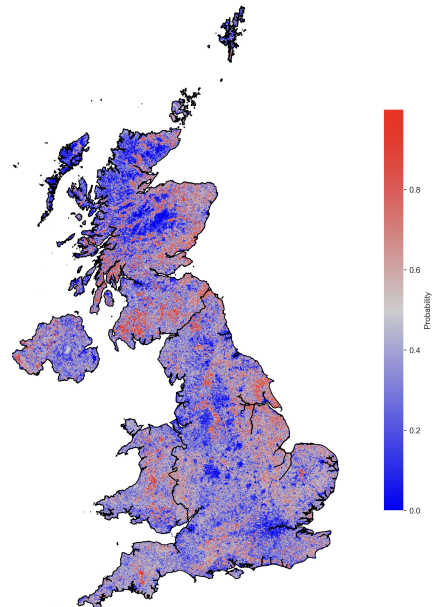
(a) *Erinaceus europaeus* (Hedgehog)



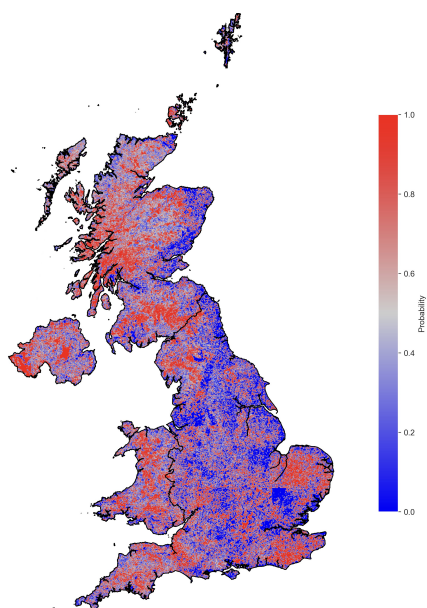
(b) *Sciurus vulgaris* (Red Squirrel)



(c) *Muscardinus avellanarius* (Hazel Dormouse)



(d) *Lepus europaeus* (Brown Hare)



(e) *Pipistrellus pygmaeus* (Soprano Pipistrelle Bat)

Figure 4.2: Habitat suitability probability maps across the UK

Chapter 5

Discussion on Results

5.1 Analysis of MaxEnt Performance

5.1.1 Trade-offs in Hyperparameter Tuning

Two separate Maxent pipelines were trained: Pipeline A (AUC tuned) and Pipeline B (Boyce Index tuned). The results clearly demonstrate the trade-offs that exist in optimizing for a single objective. Models that were optimized for Boyce Index consistently achieved high calibration scores. For example, the Boyce Index tuned model for the *Lepus europaeus* (Brown Hare) achieved a perfect Boyce Index of 1.00, and the *Muscardinus avellanarius* (Hazel Dormouse) model achieved 0.96. However, this extreme performance often severely compromised discrimination performance. The *Lepus europaeus* model, despite its perfect Boyce Index, recorded an AUC of 0.2. This highlights a critical flaw despite high calibration, often known as overfitting. Conversely, AUC tuned models generally showed better discrimination but often lower calibration than their Boyce Index tuned counterparts. For example, the *Lepus europaeus* model tuned on AUC improved the AUC to 0.44 but the Boyce Index decreased to 0.68.

5.1.2 Effectiveness of Ensembling

To mitigate the individual weaknesses of the models and take advantage of their complementary strengths, a soft-voted ensemble was implemented. The ensemble model often dominated the individual models when evaluated using the merged metric. For example, the *Sciurus vulgaris* (Red Squirrel) ensemble model achieved a merged metric score of 0.78, outperforming both the AUC tuned model (0.74) and Boyce Index tuned model (0.73). This

shows that combining the models optimized for different objectives yielded a superior predictive result.

5.2 Analysis of DeepMaxEnt performance

5.2.1 Comparative Performance & Ensemble Analysis

The DeepMaxEnt approach involved training four distinct models per species, varying the loss functions (BCEWithLogitsLoss and DeepMaxEnt Loss) and the optimization metric (AUC and Boyce Index). A detailed ensemble strategy was applied, evaluating the power set of these four models. The results confirm the superiority of the DeepMaxEnt approach. Comparing the the best MaxEnt scores with the final DeepMaxEnt scores resulted in a noticeable improvement. For example, for the *Muscardinus avellanarius*, the score increased significantly from 0.62 (MaxEnt ensemble) to 0.756 (DeepMaxEnt ensemble). Table 4.7 shows that the best models were usually the ensembles. For both the *Sciurus vulgaris* and *Muscardinus avellanarius*, the ensemble including all of the four models produced the highest mixed metric score. This suggests that each model captured unique and complementary aspects of the species' distribution.

5.2.2 Model selection and the *Pipistrellus pygmaeus* (Soprano Pipistrelle Bat) anomaly

The final model selection for inference was done primarily on the mixed metric score, aiming for the best balance between AUC and Boyce Index. However, the results for the *Pipistrellus pygmaeus* required thorough inspection. The Maxent models consistently produced higher mixed metric scores for the *Pipistrellus pygmaeus* than the DeepMaxEnt models; 0.54 for the MaxEnt ensemble and 0.378 for the DeepMaxEnt model. However, the MaxEnt scores were found to be artificially inflated by very high Boyce Index values (≈ 0.89 for the MaxEnt models) while the AUC values were severely low (≈ 0.15 for all MaxEnt models). Therefore, the decision was made to select the DeepMaxEnt model (BCEWithLogitsLoss tuned on AUC) despite the lower mixed score. This choice prioritized the model where the AUC and Boyce Index were relatively balanced, providing a more reliable prediction than a fundamentally flawed MaxEnt model.

5.3 Spatial analysis of Habitat suitability

This section addresses the primary research question: Where are the most environmentally suitable areas for the endangered species selected in the UK? The high resolution habitat suitability maps (Figure 4.2) generated by the best-performing DeepMaxEnt models provide a detailed visual answer.

5.3.1 *Erinaceus europaeus* (Hedgehog)

The *Erinaceus europaeus* shows broad suitability across much of the UK. However, there are clear limitations in upland areas. High suitability is predicted across most of England, particularly the central, eastern, and southern regions. Northern Ireland also shows significant areas of high suitability. In contrast, the primary areas of low suitability are concentrated in Wales and the upland regions, most notably the Scottish Highlands and central Scotland, likely due to climatic factors and land cover types.

5.3.2 *Sciurus vulgaris* (Red Squirrel)

The habitat suitability map for the *Sciurus vulgaris* presents the most geographically concentrated areas of suitability. The model predicts exceptionally high suitability across almost all of Scotland, including the Highlands and islands. This is expected as most of the presence points indicate that they prefer these areas. Northern Ireland and Northern England also come across as highly suitable. In contrast, the central, southern, and eastern parts show low suitability. This clear distinction highlights Scotland and Northern Ireland as the strongholds for the *Sciurus vulgaris*.

5.3.3 *Muscardinus avellanarius* (Hazel Dormouse)

The distribution for the *Muscardinus avellanarius* presents almost an inverse suitability to the *Sciurus vulgaris*. The habitat suitability map shows a strong divide between the north and the south of the UK. England and Wales are predicted to have a high suitability across most of their geographical sites, especially southwest of England, south England and along the borders of Wales. On the other hand, the model predicts extremely low suitability across most of the geographical sites in Scotland. Notably, the area of London is highlighted as very unsuitable.

This is likely driven by the high degree of urbanization, which leads to absence of connected, dense vegetation habitats essential for the species' survival. Northern Ireland seems to have generally higher suitability across its landscape but it is relatively low, with some spots being of extremely low suitability. This highlights that the conservation strategies for the *Muscardinus avellanarius* (Hazel Dormouse) must be focused primarily on maintaining and restoring suitable habitats within England and Wales.

5.3.4 *Lepus europaeus* (Brown Hare)

The *Lepus europaeus* habitat suitability map shows a more fragmented landscape. There are significant patches of high suitability which are scattered across eastern and southern England, parts of Wales, and the southern side of Scotland. The rest of the area seem to exhibit low suitability. The Scottish Highlands, the eastern side of Northern Ireland, and central England and predominantly areas of low suitability. Notably, the area of London is highlighted as very unsuitable. This patchy prediction likely highlights the *Lepus europaeus*'s preference for open agricultural landscape, which are distributed unevenly across the country.

5.3.5 *Pipistrellus pygmaeus* (Soprano Pipistrelle Bat)

The habitat suitability map for the *Pipistrellus pygmaeus* shows the most widespread distribution of high suitability among the five species studied. High probability areas are densely scattered across the geographical sites of the UK, including England, Wales, Northern Ireland, and Scotland. The most notable area of suitability is across the Lough Neagh Lake in Northern Ireland. This finding is ecologically significant, as large water bodies like Lough Neagh serve as excellent feeding habitats, supporting high densities of the insects that these bats prey on. It is important to note, however, that while critical for foraging, these aquatic environments do not provide roosting sites, for which bats require other habitats like woodland or buildings. Nevertheless, this result strongly underlines the importance of open water and wetland ecosystems for supporting this species. Some of the low suitability areas are highlighted as around eastern Scotland, north-western England around the border of Wales, eastern England and across London.

5.4 Analysis of Key Environmental Drivers

This section addresses the research question: What are the key environmental conditions that conservation efforts should prioritize for the selected endangered species in the UK? To understand the factors driving predictions of the best DeepMaxEnt models, SHAP (SHapley Additive exPlanations) analysis was utilized. The SHAP summary plots (shown in section B.4) visualize the importance of features and their impact on model output. The color indicates the feature value, where red is high and blue is low, and the X-axis position indicates the direction on the prediction (positive or negative).

5.4.1 Dominance of Land Surface Temperature

Across all the five species, Land Surface Temperature (LST) came across as the single most influential feature, indicating that the climate is the primary determinant of habitat suitability at the UK scale. However, the direction of this influence varies differently between species. LST has an overwhelming impact on the *Sciurus vulgaris* and *Lepus europaeus*. High LST values strongly decrease predicted habitat suitability, while low LST values strongly increase the predicted habitat suitability. This means that both of these species prefer cooler climates. On the other hand, the *Erinaceus europaeus*, *Muscardinus avellanarius*, and the *Pipistrellus pygmaeus* show a strong positive correlation with LST. This means that high temperatures significantly increase habitat suitability for these species.

5.4.2 Land Cover and Environmental Indices

While LST sets the influence on a broader scale, land cover and other environmental indices dictate suitability within LST viable regions: For the *Lepus europaeus*, an Arable Land Cover and Improved Grassland Land Cover are the most important features. High availability of these land types significantly increases suitability, highlighting the species' preference towards agricultural landscapes. This is further supported by the fact that the plot shows favourability towards lower Normalized Difference Water Index (NDWI) and higher Bare Soil Index (BSI), which means that this species prefers open areas with soil and avoids wet areas. For the *Erinaceus europaeus*, the model shows a positive trend between a Suburban Land Cover and probability. This suggests that suburban gardens and green spaces can come across as suitable habitats for hedgehogs. This is further supported by the fact that the plot shows favourability

towards lower Bare Soil Index (BSI) and higher towards a Improved Grassland Land Cover, which means that this species prefers green lands without open areas with soil. For the *Muscardinus avellanarius*, beyond the LST requirements, the Soil-Adjusted Vegetation Index (SAVI) is an important factor, with higher SAVI values (indicating areas with dense and healthy vegetation) contributing towards higher suitability. This highlights their need for vegetation structures. The inverse relation with Normalized Difference Snow Index (NDSI) and Modified Normalized Difference Water Index (MNDWI) further supports that this species prefers areas that are dry (little surface water) and areas with little snow cover. For the *Pipistrellus pygmaeus*, the model indicates an inverse relationship with the Improved Grassland Land Cover and a notable negative impact from proximity to roads. Meanwhile, indices related to water, such as MNDWI show a positive trend towards suitability, highlighting the importance of water bodies for this species for feeding as they skim over the water at night to feed on the aquatic insects. For the *Sciurus vulgaris*, besides the strong influence of lower LST, this species shows a negative trend between BSI and suitability. On the other hand, it shows a positive trend between the Improved Grassland Land Cover and suitability. This shows an avoidance of open soil areas and a preference for vegetation cover.

5.5 Mitigating Data Limitations with Advanced Modeling

This section addresses the research question: To what extent can advanced modeling techniques mitigate the data limitations inherent in presence-only datasets where true absences are unavailable? The findings of this dissertation demonstrate that while such limitations cannot be eliminated entirely, a multi-faceted approach combining bias handling, advanced deep learning, and robust evaluation can significantly mitigate their impact. The foundational challenge of presence-only data is the lack of true absences, which is compounded by sampling bias in occurrence records. The implementation of TGB was the first and most critical step. The models were trained on a dataset that better reflects the sampling effort and environmental space relevant to the target species. The DeepMaxEnt models consistently and significantly outperformed the traditional MaxEnt approach, due to using deep neural networks to learn relationships that simpler models cannot capture. This capacity is particularly useful when dealing with incomplete and noisy data. Furthermore, the use of ensembling proved to be a powerful mitigation strategy. By averaging the predictions of multiple models, the final

output becomes more robust. Finally, mitigating data limitations also requires a more sophisticated approach to evaluation. Relying on a single metric is risky. The development and use of a merged metric, which balances discrimination (AUC) and calibration (Boyce Index), provided a more trustworthy assessment of performance.

5.6 From Models to Management: A Science-for-Policy Perspective

For scientific models to be effective in policy, they must be trusted by decision-makers. This research prioritized this by creating interpretable decision-support tools, not just predictive models. The use of SHAP analysis was critical, transforming the models from black-boxes into transparent instruments by explaining *why* an area is predicted as suitable. For example, knowing a woodland is suitable for the Hazel Dormouse due to its dense vegetation provides conservation officers with direct, defensible evidence for planning consultations. With this trust established, the habitat suitability maps serve as a starting point for strategic action. They allow organizations to more effectively target limited resources for monitoring and funding, justify specific land management interventions, and provide evidence for protecting critical habitats during planning processes. By presenting not just predictions, but the reasoning behind them, this research empowers decision-makers to act more strategically to protect the UK's endangered species.

Chapter 6

Project Management

6.1 Project Planning and Gantt Chart

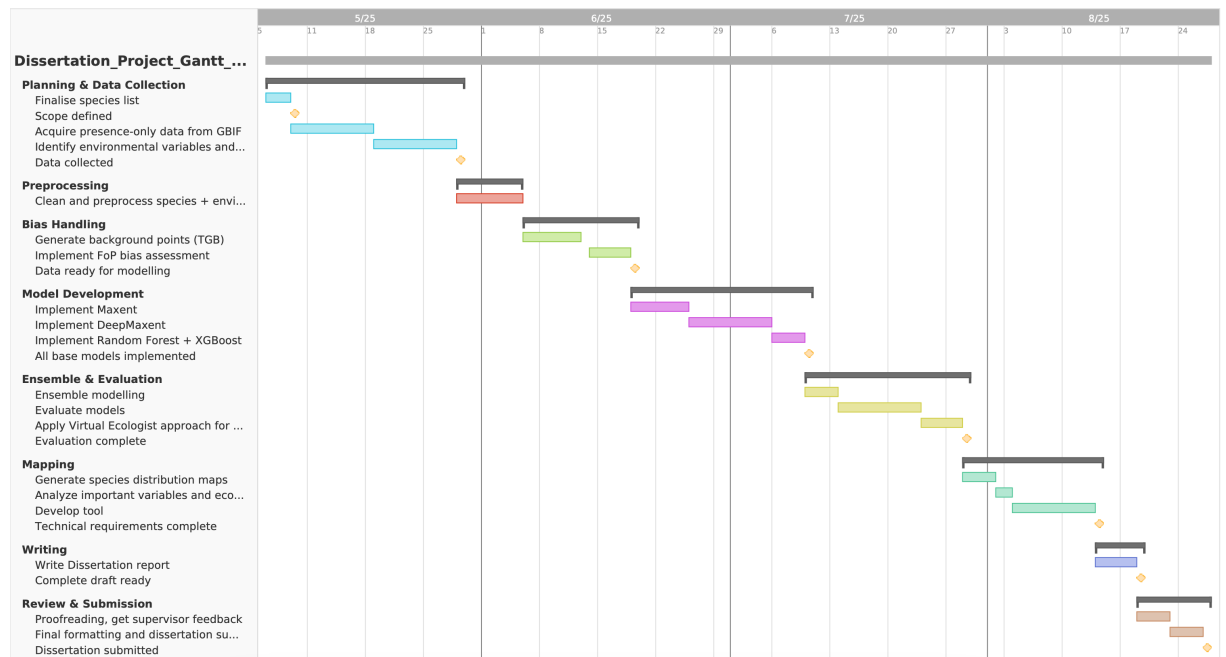


Figure 6.1: Gantt Chart

6.2 Tools and Technologies Used

The tools used were as follows: (1) GBIF API for presence data collection. (2) OpenStreetMap API and OSMnx for road data collection. (3) Python for the overall implementation of the project. (4) Pandas and Geopandas for data handling. (5) Numpy for mathematical operations on the dataset. (6) Matplotlib & seaborn for visualization and maps. (7) Sci-kit learn for encoding, scaling and modeling the data for MaxEnt. (8) Pytorch for modeling DeepMaxEnt.

(9) SHAP for understanding feature importances of models. (10) BallTree for efficient nearest-neighbour searching when engineering proximity-based features. (11) TQDM for loading bars during processing of data. (12) Pyproj for transforming between coordinate systems. (13) Rasterio for dealing with rasters. (14) Joblib and pickle for saving models and scalars. (15) Scipy for interpolation for creating output maps. (16) Shapely for reading the UK shapefile. (17) EarthEngine API for sourcing environmental variables. (18) Datetime for handling dates in data.

6.3 Risks & Mitigation Strategies

6.3.1 Data Unavailability

Initially, species from the IUCN Red List were chosen but it was found that some data was unavailable for the species listed there. Hence, the strategy was to pivot and select species from the UK BAP Priority List, which had sufficient data for the analysis to proceed.

6.3.2 Technical Complexity

DeepMaxEnt was a new concept that came out recently in 2025. This presented a risk of encountering challenges in understanding it. The strategy was to seek expert collaboration and guidance from its author.

6.3.3 Computational Inefficiency

Processing large geospatial datasets is computationally expensive, posing delays. For example, engineering the proximity feature by naively comparing every point to every other point was very slow. Hence, the strategy was to use efficient algorithms, such as Ball Tree.

6.3.4 Uncertain results

This was a complex project that required dealing with presence-only data and required extensive data collection, process designing, modeling and visualization. This ran the risk of becoming disorganized. The strategy used was to stick to a clear workflow guided by the project plan and using modular, reusable code files wherever possible and making clear

pipelines and scripts for the processes. This ensured any errors were rapidly solved and made the codebase easier to maintain.

6.4 Deviations from Initial Plan

Apart from deviations in the timeline (early completion or late completion according to the Gantt chart), there were some modifications to the initial plan. The idea to use FoP bias assessment was dropped due to it requiring true absences, which is not possible in this project case. Moreover, Random Forest and XGBoost were not implemented as they are not suitable for predicting suitability when it comes to presence-only datasets. A Virtual Ecologist (Zurell et al., 2010) approach for evaluating the outputs was considered, which simulates ecological systems and observation processes. However, it also requires the true distribution of the species to be known, which needs absences. Therefore, this idea was dropped. The writing of the report was initially saved for the end of the timeline. However, upon discussion with my supervisor, it was agreed to do it side by side with the project implementation. Although this caused a delay in the overall timeline due to working on two things at once, it was an important step to make sure no processes were undocumented. Instead of developing an open-source tool due to time constraints, habitat suitability maps for the UK were created which can be shared in an open source manner for conservationists to access. Apart from these deviations in sub-stages, all the stages of the plan were followed sequentially and completed.

6.5 Initial Topic Selection and Rationale

The final dissertation topic was selected after considering several potential topics that leverage geospatial data for environmental and conservation purposes. This section outlines the initial candidate topics and provides the rationale for the final selection.

6.5.1 Candidate Topics

Urban Heat Island Effect Analysis

This topic proposed mapping urban expansion in city like London to identify areas where rising temperatures could be mitigated with green infrastructure.

1. Pro: High relevance for UK urban planning and climate resilience.
2. Con: Thermal resolution of satellite imagery may be too low to differentiate heat from rooftops with heat from major sources such as traffic, industrial processes, etc.

Biodiversity Loss Assessment

A topic focused on analyzing habitat fragmentation from land-use changes to identify regions at risk of species decline.

1. Pro: Directly addresses a critical conservation issue.
2. Con: The scope was potentially too broad. Quantifying biodiversity from satellite data alone is complex and abstract. The current dissertation refines this idea into a more focused and methodologically robust project.

Blue Carbon Potential in UK Seagrass Meadows

This topic focused on mapping seagrass restoration potential in Wales and Scotland, supporting national conservation goals to recover this critical habitat.

1. Pro: Addresses a significant UK-specific conservation failure (92% seagrass meadows lost) and supports ongoing initiatives.
2. Con: Remote sensing of submerged vegetation is technically difficult due to factors like water clarity, depth, etc.

Invasive Species Monitoring (Japanese Knotweed)

This topic aimed to create an early-alert system for councils by detecting the unique seasonal growth patterns of Japanese Knotweed.

1. Pro: High economic relevance for the UK, where the species causes an estimated £166 million in damage annually.
2. Con: Isolating the spectral signature of a single plant species at a medium resolution from satellite data is very challenging, especially when it grows among other vegetation.

Ultimately, the current topic of Species Distribution Modeling was chosen as it provided the best balance of UK conservation relevance, data availability, and methodological innovation. It narrowed the broad "Biodiversity Loss" idea into a concrete, achievable project with clear applications for protecting specific, endangered species.

Chapter 7

Practical Realisation

This project was realized as an action research study, centered on the development and experimental application of a methodology to address a conservation problem. The core of this work was to produce a solution with actionable insights in the form of suitability maps to support conservation planning for endangered species in the UK. The entire project was conducted with full regard for its ethical implications.

7.1 Development of Methodology

The primary step in this action research project was the creation of a data processing and modeling pipeline. this pipeline served as the primary tool for investigating the research questions. The methodology was designed to use traditional SDMs (MaxEnt) with advanced deep learning approaches (DeepMaxEnt) using presence-only data. This involved sourcing species occurrence data from GBIF and a wide array of environmental variables from trusted repositories like Google Earth Engine and UKCEH. A modular pipeline was implemented in Python. Key components included modules for data ingestion, cleaning, feature engineering and preprocessing. This systematic approach ensured the process was reusable and robust in case of any changes or errors. The modeling tools developed included custom implementations for training and tuning both MaxEnt and DeepMaxEnt models. An ensembling approach using soft voting was developed to combine the predictive strengths of multiple models. This iterative development allowed for direct comparison between different modeling strategies..

7.2 Experimental Application and Evaluation

The developed methodology and modeling tools were experimentally applied to the selected species. This phase refers to the action component of the research, where the tools were used to generate and test predictions. A key part of the methodology was the application of TGB. This mitigated the inherent sampling bias in the presence-only datasets. The project involved several prototype iterations. Initially, the MaxEnt models were trained and tuned under different objective functions. The evaluation of these initial models revealed trade-offs in performance, leading to the development of an ensemble model. Finally, DeepMaxEnt models were developed and subjected to more exhaustive ensemble experiments, representing a second iteration. This iterative process was crucial, as it incorporated a 'human-in-the-loop' validation step. An early version of the *Pipistrellus pygmaeus* model, for example, produced a map showing 100% suitability exclusively over Lough Neagh, with 0% suitability across the rest of the entire UK. A manual sanity check of this map, combined with an inspection of the corresponding SHAP plot which revealed the model had overfit substantially on a single water-related index, allowed for this bias to be identified and rectified. This highlighted the importance of not just relying on quantitative metrics but also on qualitative, expert-driven assessment of the model outputs.

The primary end-product of these project lifestyle activities was the set of five high-resolution habitat suitability maps. The validity of these maps was assessed through a testing plan where all models were evaluated against a 20% validation dataset. The test results, based on AUC, Boyce Index, and the innovative merged metric score provided a quantitative measure of the solution's effectiveness and guided the selection of the final models for each species. The final output of this practical realisation is a validated set of modeling tools and a corresponding methodology that successfully generated actionable habitat suitability intelligence from incomplete data, fulfilling the objective of the project.

Chapter 8

Evaluation and Reflection

8.1 Comparison of Outcomes with Objectives

The project set out to address three primary research questions identified in Chapter 3. The evidence presented in Chapter 5 confirms that these objectives were successfully met.

8.2 Reflection on Processes and Methods

A reflection on the methods and tools employed revealed both strengths and areas for learning. The methodology allowed for an iterative process where initial modeling attempts (MaxEnt) provided insights that directly informed the development of a more advanced second iteration (DeepMaxEnt). This cycle of act → evaluate → reflect was crucial, particularly in identifying and overcoming challenges like the anomaly discovered with the *Pipistrellus pygmaeus* model (See Section 5.2.2) where a simplistic evaluation would have led to the selection of a flawed model. This insight reinforces the need of using a balanced combination of metrics to get a holistic view of model performance. Similarly, another lesson learned was that data quality is dominant over model complexity. While the DeepMaxEnt model performed the best, its success was built upon carefully engineered and cleaned data. This project restablished the idea that complex algorithms alone cannot compensate for poor data.

The use of a Ball Tree for proximity analysis was computationally efficient and scaled well to large datasets. Similarly, the OSMnx library provided a reliable method for integrating road data. However, the data ingestion and cleaning phase was the most time-consuming part. The lesson learned here is that even with powerful tools, the quality of the final product is

dependent on extensive and often manual data preparation.

The habitat suitability map produced are intended for use by ecologists and conservationists. A limitation of this project was that no formal usability study with the end-users was conducted for true feedback.

On a personal level, this project served as a significant learning experience. A key strength was my technical proficiency in Python and its data science and machine learning libraries, which was essential for implementing the complex data processing and modeling pipelines. However, a weakness was initially underestimating the sheer amount of time required for data sourcing, cleaning, and feature engineering, which proved to be the most time-consuming phase of the project. From a project management perspective adhering to the Gantt chart was largely successful. The most important lesson learned was the value of adaptability; being willing to deviate from the initial plan by dropping unfeasible methods was critical to the project's successful completion.

8.3 Future Work

Future work could include actually using these maps to go to highly suitable areas for species and conduct an investigation with conservation professionals on whether the species are actually found there or not; and if not, why not? Moreover, since the maps are for a specific date (1 May 2025, as mentioned in Section 4.7), a next step would be to predict habitat suitability in the winter to see how suitability might shift. Furthermore, future iterations could also include other influential variables, such as human population density or habitat connectivity to improve model performance. Moreover, the habitat suitability maps produced are inherently compatible with Geographic Information Systems (GIS). GIS are frameworks for capturing, managing, and analyzing spatial and geographic data. The final raster maps from this project could be easily overlaid as a new data layer onto existing GIS maps containing administrative boundaries, such as protected areas or local planning zones. This would be incredibly useful for conservation planners, as it would allow them to quickly identify highly suitable habitats that currently lack legal protection or highlight areas where proposed development projects could threaten key strongholds for these endangered species. Finally, the use of species simulation following the work of Garzon-Lopez et al. (2016) can be considered as potential future work for testing the SDMs.

8.4 Ethical Implications

Though approved, licensed and publicly available data was used, there are other ethical implications to reflect on according to the AI4People framework (Floridi et al., 2018).

Beneficence: By developing accurate and reliable species distribution maps, this work aims to promote the well-being of endangered species and support the health of the UK ecosystem.

This maximizes the impact of limited conservation resources.

Non-maleficence: A critical issue is the potential for harm. If the suitability maps were made available in an unrestricted manner, they could be misused for illegal activities such as poaching or for planning developments that disturb habitats. Therefore, the availability of the final maps would require careful consideration, like sharing them directly with trusted conservation bodies.

Justice: Sharing the results with conservationists promotes fairness by ensuring the benefits of the research can be used for the common good, rather than being commercialized.

Explicability: For any AI tool to be useful, its outputs must be trusted. To address the need for the explicability, the project went beyond simply generating predictions. The SHAP analysis was purposefully done to explain why the model predicts a certain area is suitable. This makes the project transparent.

Chapter 9

Conclusion

This dissertation addressed the critical challenge of modeling the habitat suitability of endangered species in the United Kingdom using presence-only data. Keeping in mind the difficulties posed by inherent sampling bias and the limitations of traditional Species Distribution Modeling, this research developed and applied a robust methodology utilizing advanced computational techniques. The study focused on five species: *Erinaceus europaeus*, *Sciurus vulgaris*, *Muscardinus avellanarius*, *Lepus europaeus*, *Pipistrellus pygmaeus*. The methodology successfully collected and consolidated diverse data sources, including species occurrence records from GBIF and environmental variables from Google Earth Engine and the UKCEF. Key steps included data cleaning, feature engineering, preprocessing, bias handling and modeling. The final step was to produce habitat suitability maps. The research involved systematic modeling using the established MaxEnt approach and the novel DeepMaxEnt approach. Furthermore, exhaustive ensemble modeling strategies via soft voting were incorporated to take advantage of the individual strengths of the models optimized under different objectives. Through evaluation using AUC, Boyce Index, and a merged metric designed to balance discrimination and calibration, the study showed the superior performance of DeepMaxEnt, particularly when utilized with ensembles. The results confirmed that combining the models optimized for different objectives results in more reliable predictions than single models.

The research successfully answered the three primary research questions: Where are the most environmentally suitable areas for the endangered species selected in the UK? What are the key environmental conditions that conservation efforts should prioritize for the selected endangered species in the UK? To what extent can advanced modeling techniques mitigate the

data limitations inherent in presence-only datasets where true absences are unavailable?

The first one was addressed by the project delivering high-resolution habitat suitability maps that provide actionable intelligence. The analysis revealed that suitability is species-dependent. Scotland and Northern Ireland appeared as strongholds for *Sciurus vulgaris*. In contrast, England and Wales provide primary suitability for *Muscardinus avellanarius* and *Erinaceus europaeus*, particularly in the south and southwest. *Lepus europaeus* suitability was more fragmented, concentrating in eastern and southern England, while *Pipistrellus pygmaeus* showed broadest suitability across the UK. The second question was addressed by SHAP analysis which identified key environmental factors which influence model predictions. Land Surface Temperature was the single most influential feature across all species, indicating the importance of climate in the UK. *Sciurus vulgaris* and *Lepus europaeus* preferred cooler climates, while the other three species preferred warmer climates. Within climatic suitability, specific land cover types and spectral indices drove suitability, such as the preference for agricultural landscapes by *Lepus europaeus*, suburban green spaces by *Erinaceus europaeus*, and dense vegetation for *Muscardinus avellanarius*. The third question was answered by showing that a multi-faceted approach, combining bias-handling, the superior learning capacity of deep learning models, the stabilizing effect of ensembling, and a robust dual-metric evaluation, can effectively mitigate the impacts of sampling bias and the lack of true absences. This dissertation makes significant contributions to both ecological conservation and computational literature. Academically, it validates the effectiveness of DeepMaxEnt and ensemble modeling in overcoming the challenges of presence-only data. Practically, the generated maps offer valuable insights to conservationists in the UK, guiding and focusing their protection efforts in habitat restoration and species reintroduction. Crucially, it demonstrates how to make complex science actionable for policy. By prioritizing model interpretability through SHAP analysis and producing clear visual aids, this work delivers a trusted evidence base that can be used to target funding, inform land-use planning, and justify conservation actions. Ultimately, it provides a framework for how advanced modeling can be effectively translated into tangible policy and outcomes for protecting the UK's biodiversity. Future work should prioritize the ground-truthing of these maps in collaboration with conservation professionals. Additionally, future research should explore seasonal variations in habitat suitability and incorporate other variables such as human population density. In conclusion, this research successfully demonstrates how deep learning and bias-mitigation

strategies can transform incomplete biodiversity data into actionable insights. It provides a framework for future SDM studies. the findings offer clear understanding of the habitat needs of key endangered species, supporting efforts to protect the UK's biodiversity in the face of ongoing environmental change.

Appendix A

Equations

A.1 Standard Scaler

$$x'_i = \frac{x_i - \mu}{\sigma}$$

where:

x_i = the original value of the i^{th} observation

μ = the mean of the feature (average of all x_i values)

σ = the standard deviation of the feature (spread of the data)

x'_i = the standardized value

A.2 Multiple Linear Regression and Logistic Regression

Multiple Linear Regression

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_n X_n + \varepsilon$$

where:

Y = the predicted value

β_0 = the y-intercept

β_1 = the regression coefficient of the first independent variable

X_1 = the first independent variable

β_2 = the regression coefficient of the second independent variable

X_2 = the second independent variable

β_n = the regression coefficient of the last (n^{th}) independent variable

X_n = the last (n^{th}) independent variable

Logistic Regression

$$\log\left(\frac{p(\mathbf{x})}{1-p(\mathbf{x})}\right) = \beta_0 + \beta_1 x_1 + \cdots + \beta_{p-1} x_{p-1}$$

A.3 Poisson Loss

$$\mathcal{L} = \sum_{i=1}^n (\hat{y}_i - y_i \log \hat{y}_i + \log(y_i!))$$

where:

n = number of data points

y_i = actual observed count for the i^{th} data point

$\hat{y}_i$ = predicted count for the i^{th} data point

$y_i!$ = factorial of the observed count

A.4 Binary Cross-entropy with Logits Loss

$$\mathcal{L} = \frac{1}{n} \sum_{i=1}^n \left[\max(z_i, 0) - z_i y_i + \log(1 + e^{-|z_i|}) \right]$$

where:

n = number of samples

y_i = target label (0 or 1) for sample i

z_i = logit predicted by the model for sample i

A.5 DeepMaxEnt Loss

$$\mathcal{L} = -\frac{1}{n} \sum_{i=1}^n \left[y_i \log(\sigma(z_i) + \epsilon) + (1 - y_i) \log(1 - \sigma(z_i) + \epsilon) \right]$$

where:

n = number of samples

y_i = target label (0 or 1) for sample i

z_i = logit predicted by the model for sample i

$\sigma(z_i)$ = sigmoid of the logit: $\frac{1}{1 + \exp -z_i}$

ϵ = small constant to avoid $\log 0$ (e.g., 1×10^{-8})

A.6 Soft Voting

$$\hat{p}(y = c) = \frac{1}{M} \sum_{m=1}^M \hat{p}_m(y = c)$$

where:

M = number of models

$\hat{p}_m(y = c)$ = probability predicted by model m for class c

The final class prediction is given as:

$$\hat{y} = \begin{cases} 1 & \text{if } \hat{p} \geq \text{threshold} \\ 0 & \text{otherwise} \end{cases}$$

A.7 Sensitivity and Specificity

$$sensitivity = \frac{TP}{TP+FN}, specificity = \frac{TN}{TN+FP}$$

where:

TP = number of positive samples correctly classified as positive

FP = number of negative samples incorrectly classified as positive

TN = number of negative samples correctly classified as negative

FN = number of positive samples incorrectly classified as negative

A.8 Pearson Correlation Coefficient for Boyce Index

$$\text{Boyce Index} = r = \frac{\sum_{i=1}^n (m_i - \bar{m})(R_i - \bar{R})}{\sqrt{\sum_{i=1}^n (m_i - \bar{m})^2 \sum_{i=1}^n (R_i - \bar{R})^2}}$$

where:

m_i = midpoint of bin i (predicted suitability)

R_i = P/E ratio for bin i

$\bar{m}$ = mean of all midpoints

$\bar{R}$ = mean of all P/E ratios

n = number of valid bins

A.9 Maximum Likelihood of Logistic Regression

$$p(y | x) = \frac{1}{Z(x)} \exp\left(\sum_{i=1}^m \lambda_i f_i(x, y)\right)$$

where:

$$Z(x) = \sum_{y' \in \mathcal{Y}} \exp\left(\sum_{i=1}^m \lambda_i f_i(x, y')\right)$$

Symbol	Meaning
x	input / context (observed instance)
y	output / label (discrete)
$\mathcal{Y}$	set of all possible labels y
$p(y x)$	model conditional probability of y given x
m	number of feature functions
$f_i(x, y)$	feature function i (often an indicator; $\mathbb{R}$ -valued allowed)
λ_i	parameter / weight (Lagrange multiplier) for feature f_i
$\sum_{i=1}^m \lambda_i f_i(x, y)$	unnormalized log-score (model score)
$Z(x)$	partition function / normalizer: $Z(x) = \sum_{y' \in \mathcal{Y}} \exp(\sum_i \lambda_i f_i(x, y'))$
y'	dummy variable running over labels in the sum for $Z(x)$
$\boldsymbol{\lambda}, \mathbf{f}(x, y)$	vector forms of λ_i and $f_i(x, y)$

Table A.1: Notation for the equation

A.10 Probability using Logistic Regression

$$p(\mathbf{x}_i) = P[Y_i = 1 \mid \mathbf{X}_i = \mathbf{x}_i] = \frac{e^{\beta_0 + \beta_1 x_{i1} + \dots + \beta_{p-1} x_{i(p-1)}}}{1 + e^{\beta_0 + \beta_1 x_{i1} + \dots + \beta_{p-1} x_{i(p-1)}}}$$

Appendix B

Figures

B.1 Linear (Ordinary) Regression vs Logistic Regression

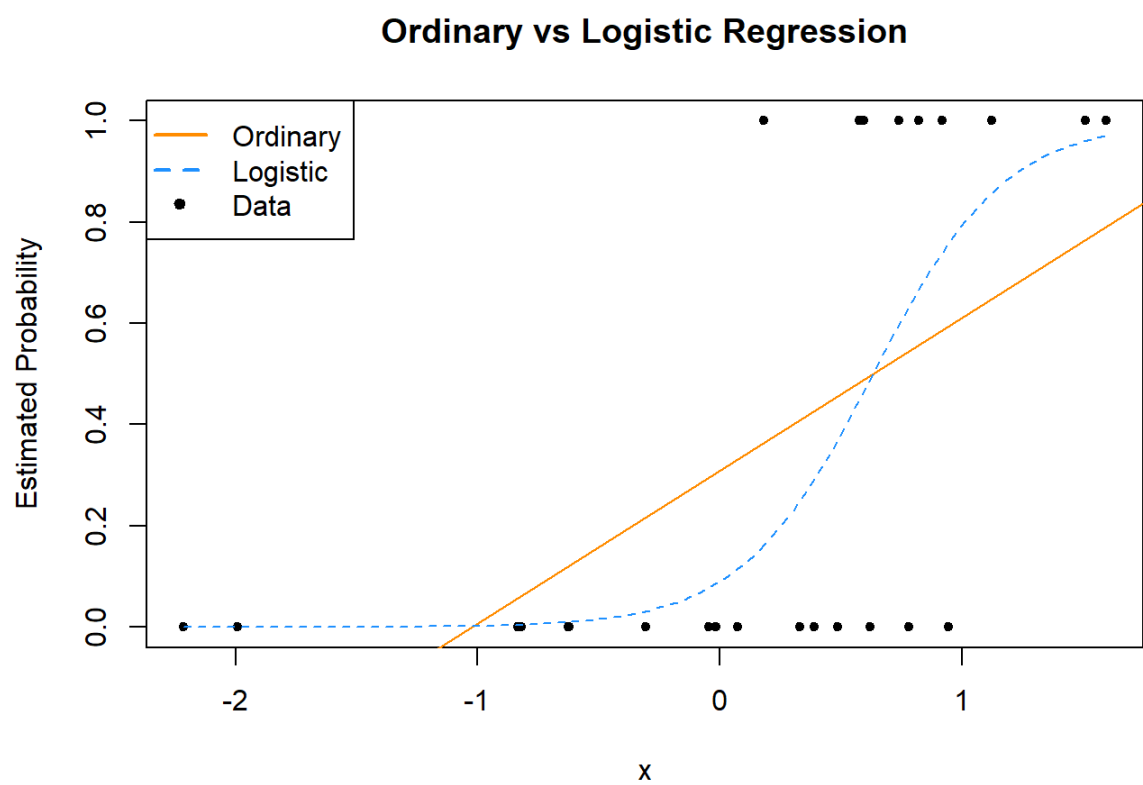
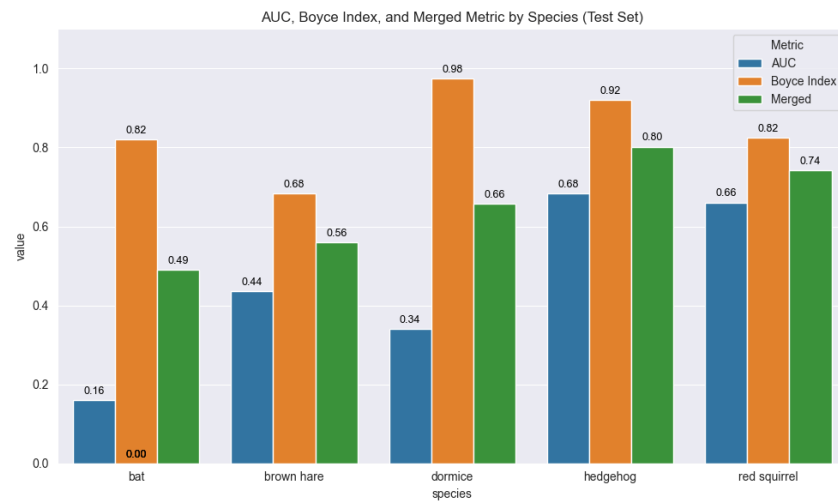


Figure B.1: Comparison of Linear vs Logistic regression

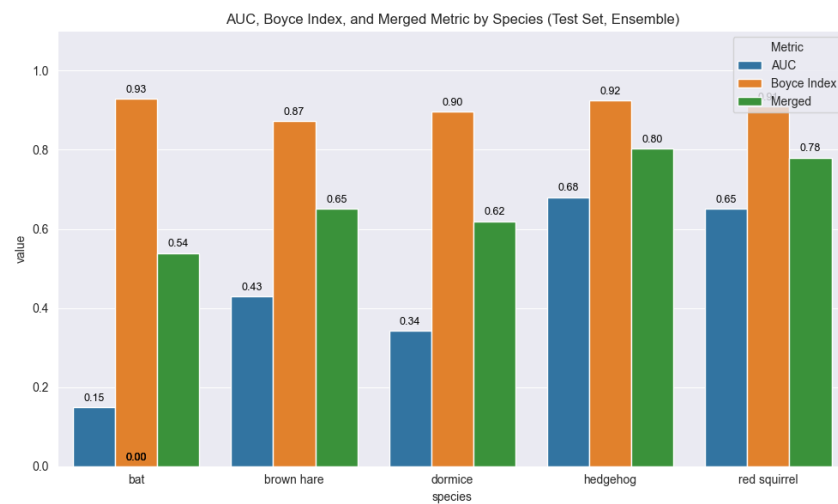
B.2 MaxEnt model performances



(a) MaxEnt models tuned on Boyce Index



(b) MaxEnt models tuned on AUC



(c) Ensemble MaxEnt model

Figure B.2: Comparison of MaxEnt models for habitat suitability

B.3 DeepMaxEnt ensemble results

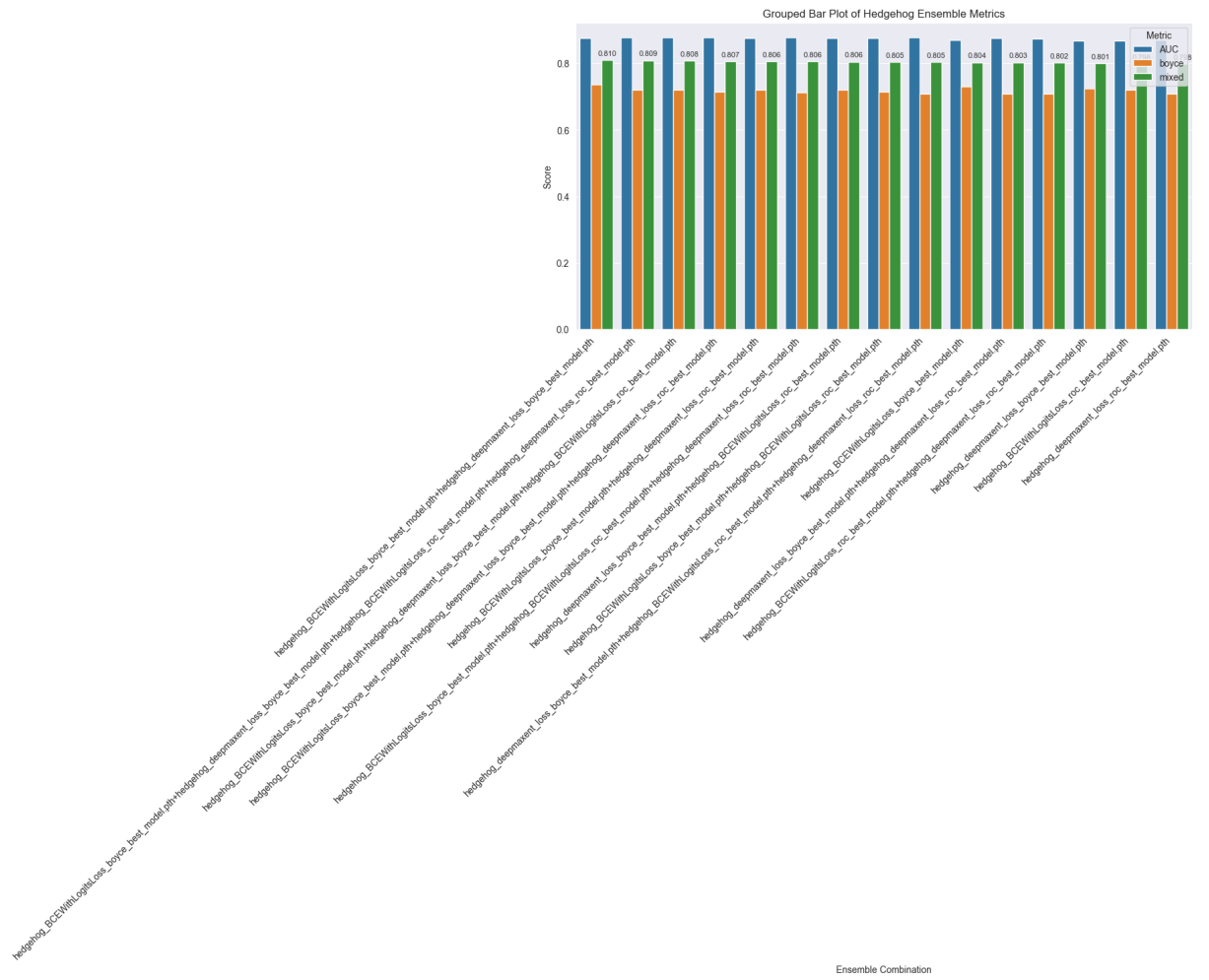


Figure B.3: *Erinaceus europaeus* (Hedgehog)

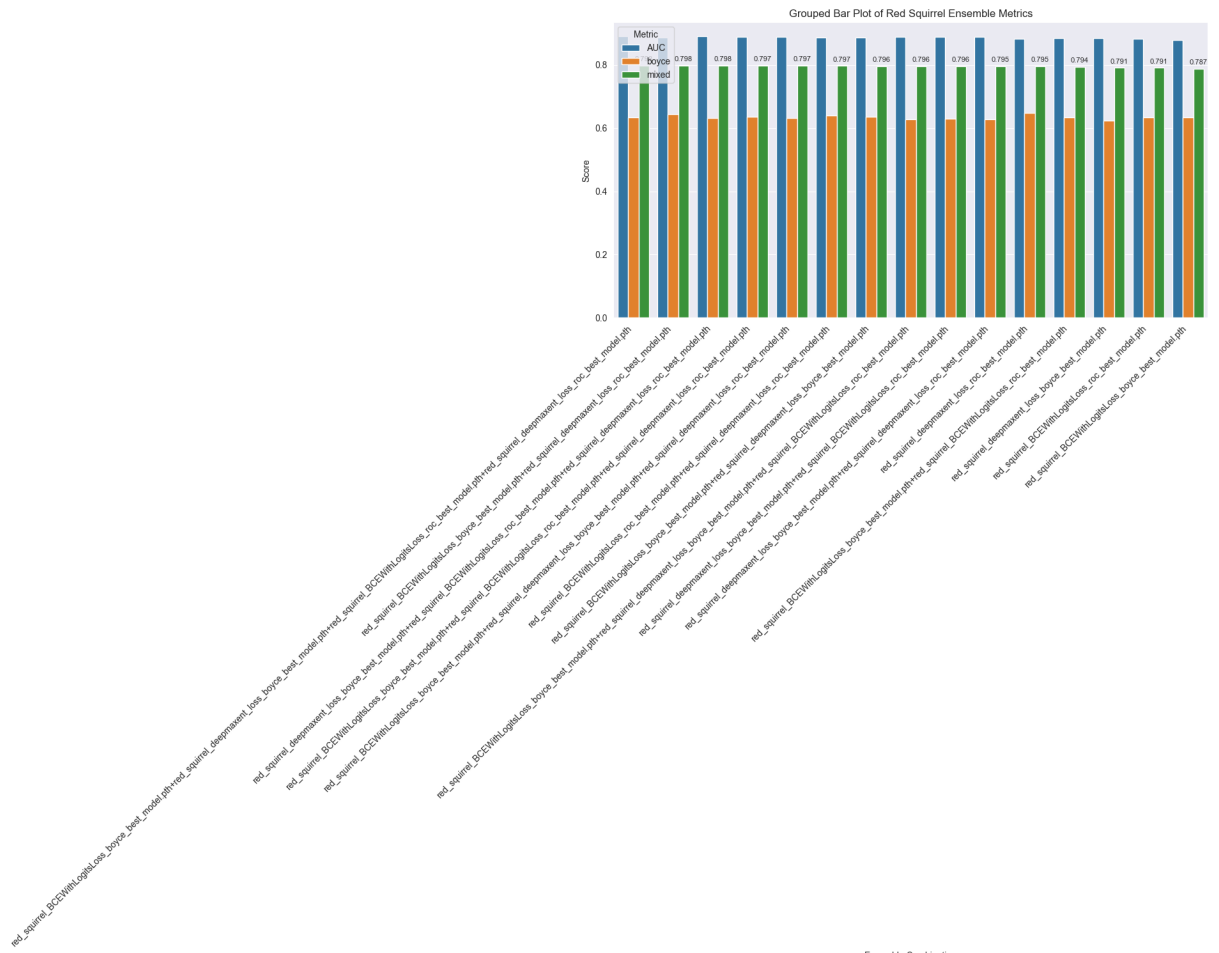


Figure B.4: Sciurus vulgaris (Red Squirrel)

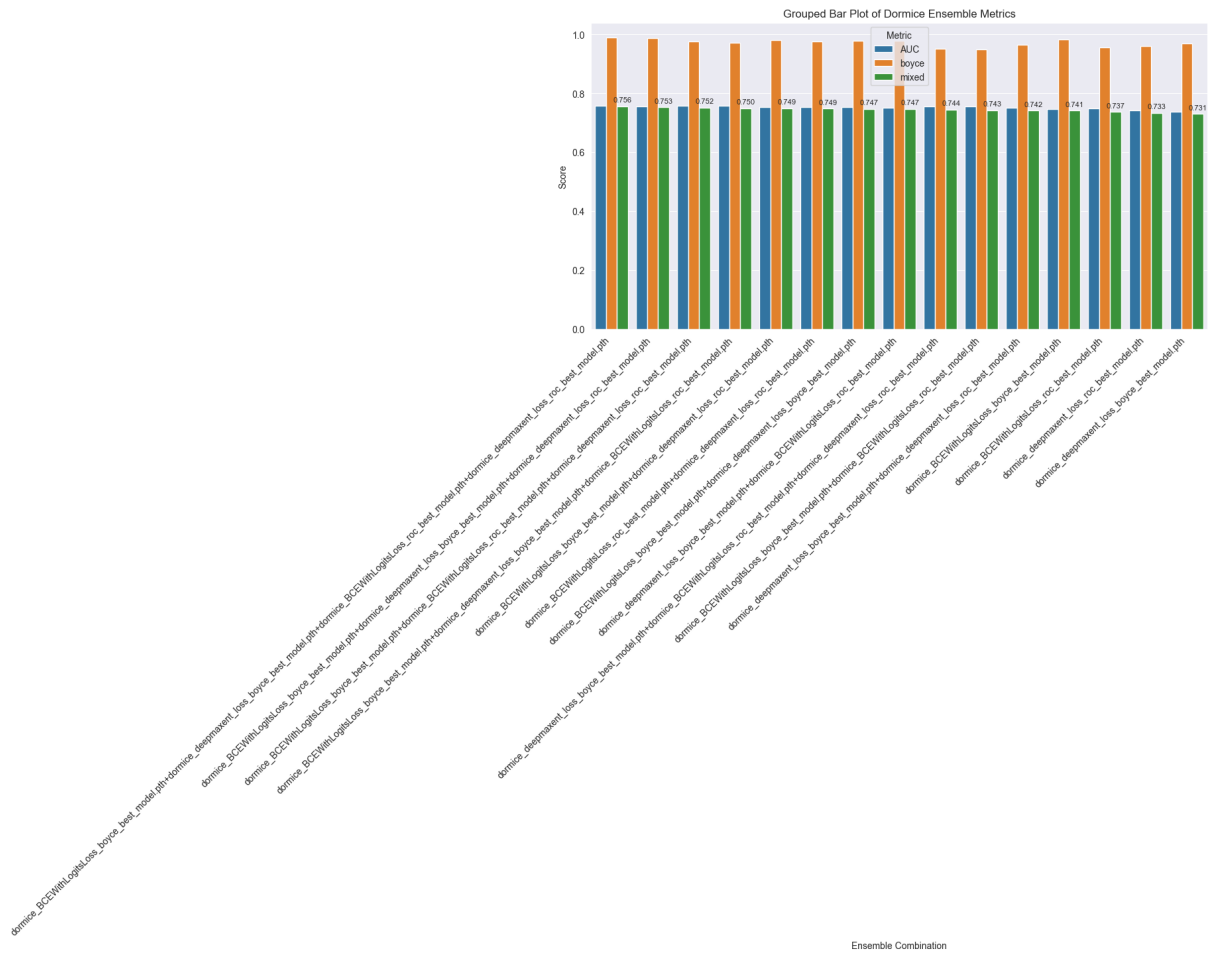


Figure B.5: Muscardinus avellanarius (Dormice)

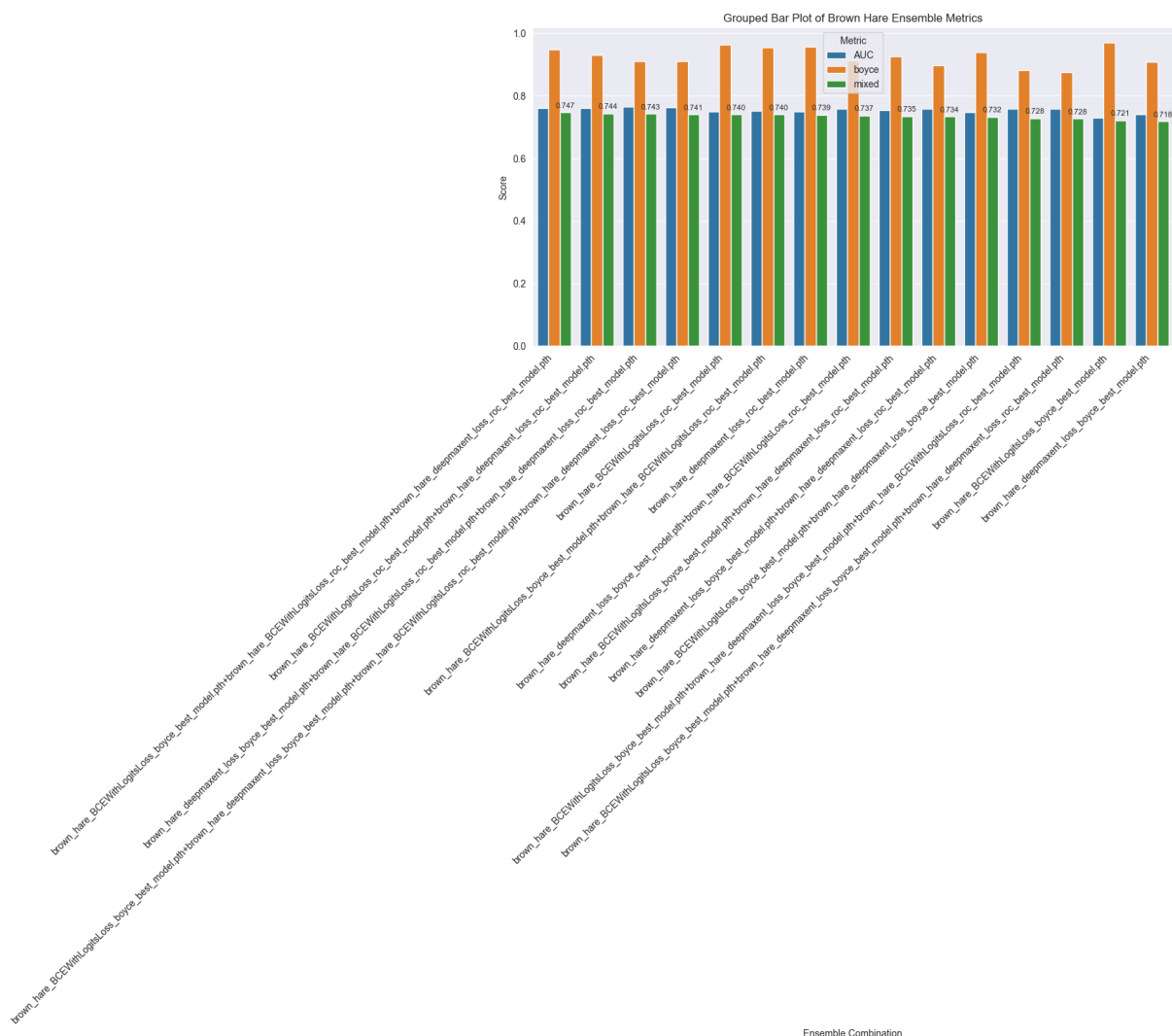


Figure B.6: *Lepus europaeus* (Brown Hare)

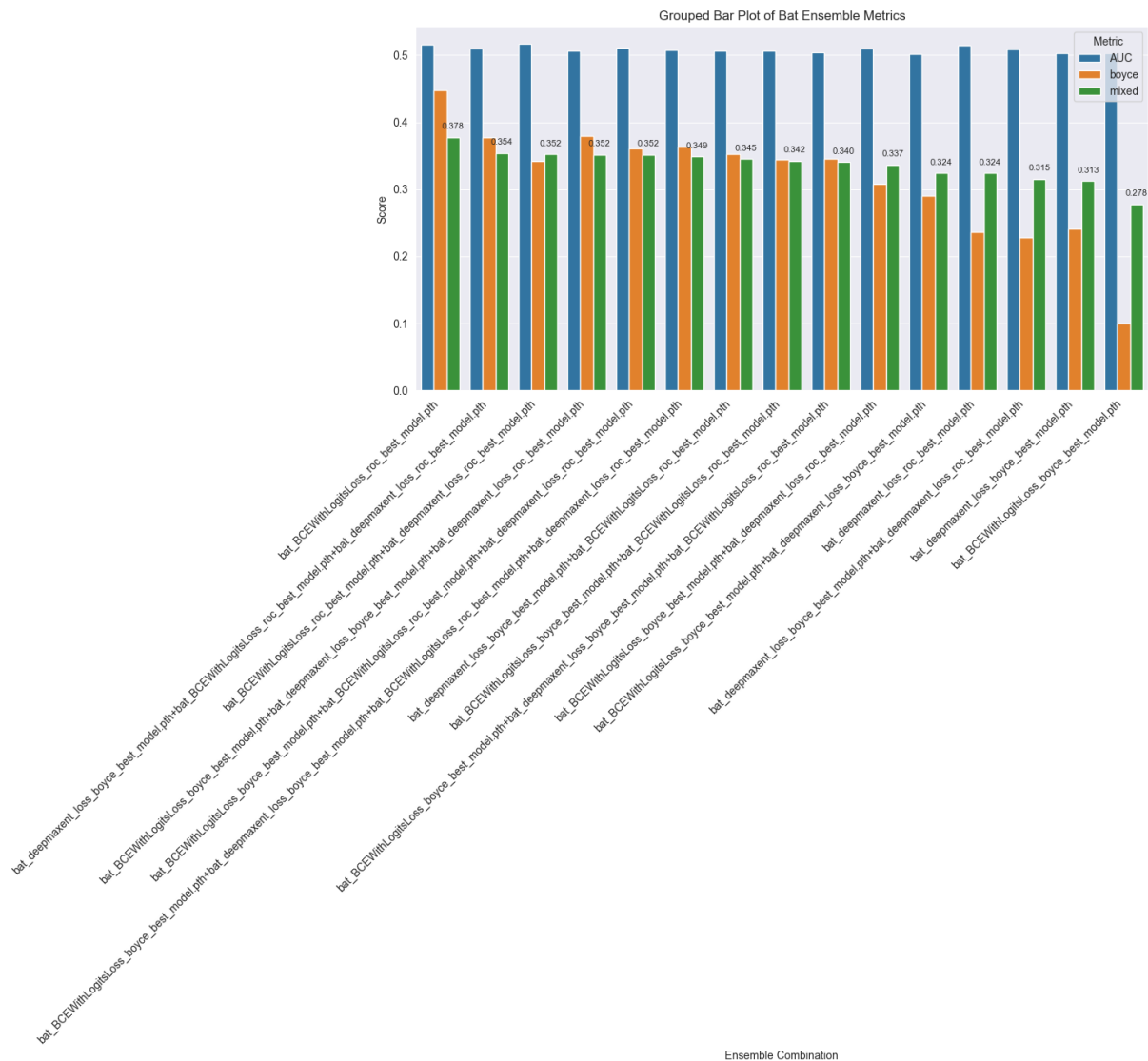


Figure B.7: *Pipistrellus pygmaeus* (Bat)

B.4 SHAP plots

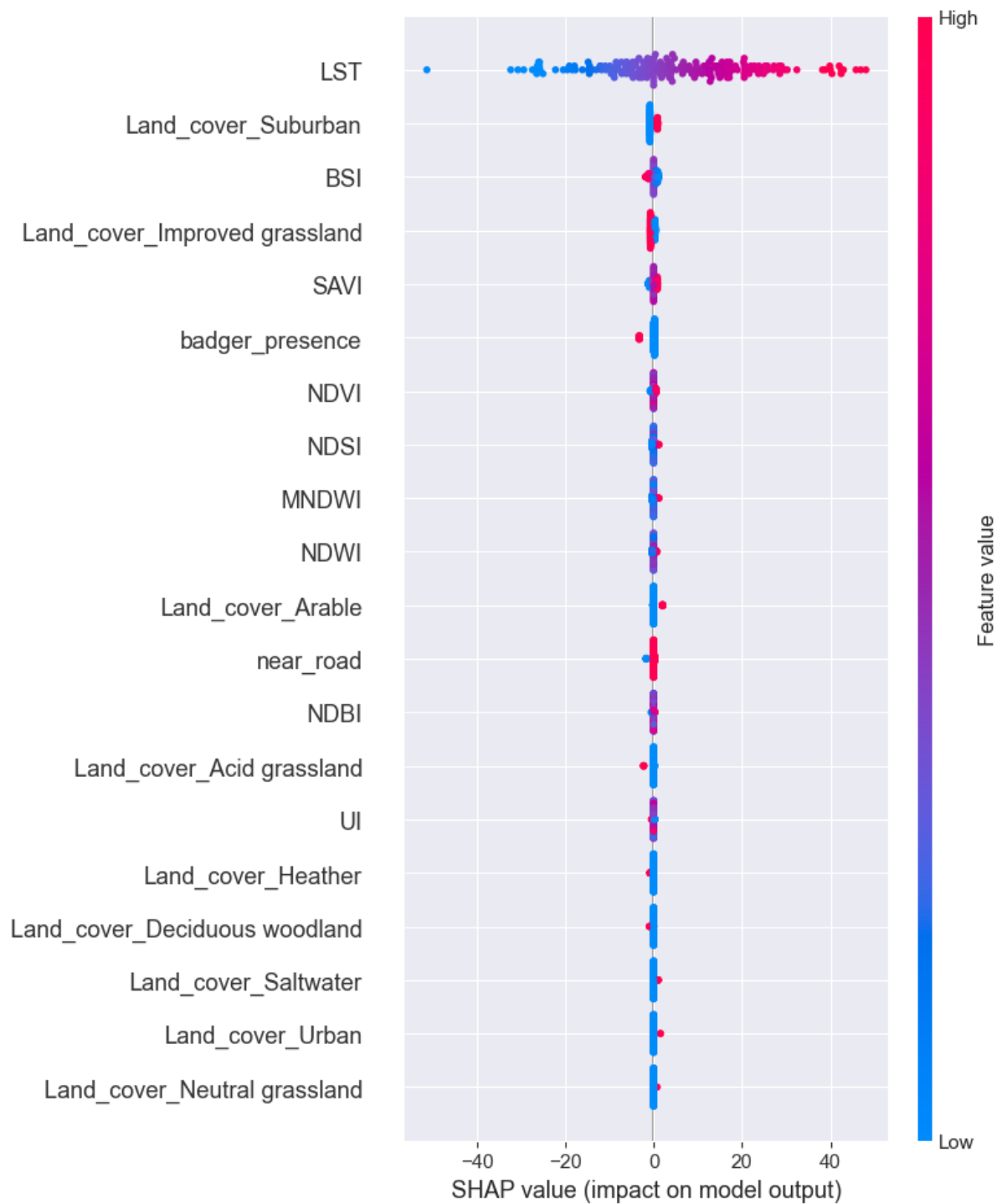


Figure B.8: *Erinaceus europaeus* (Hedgehog)

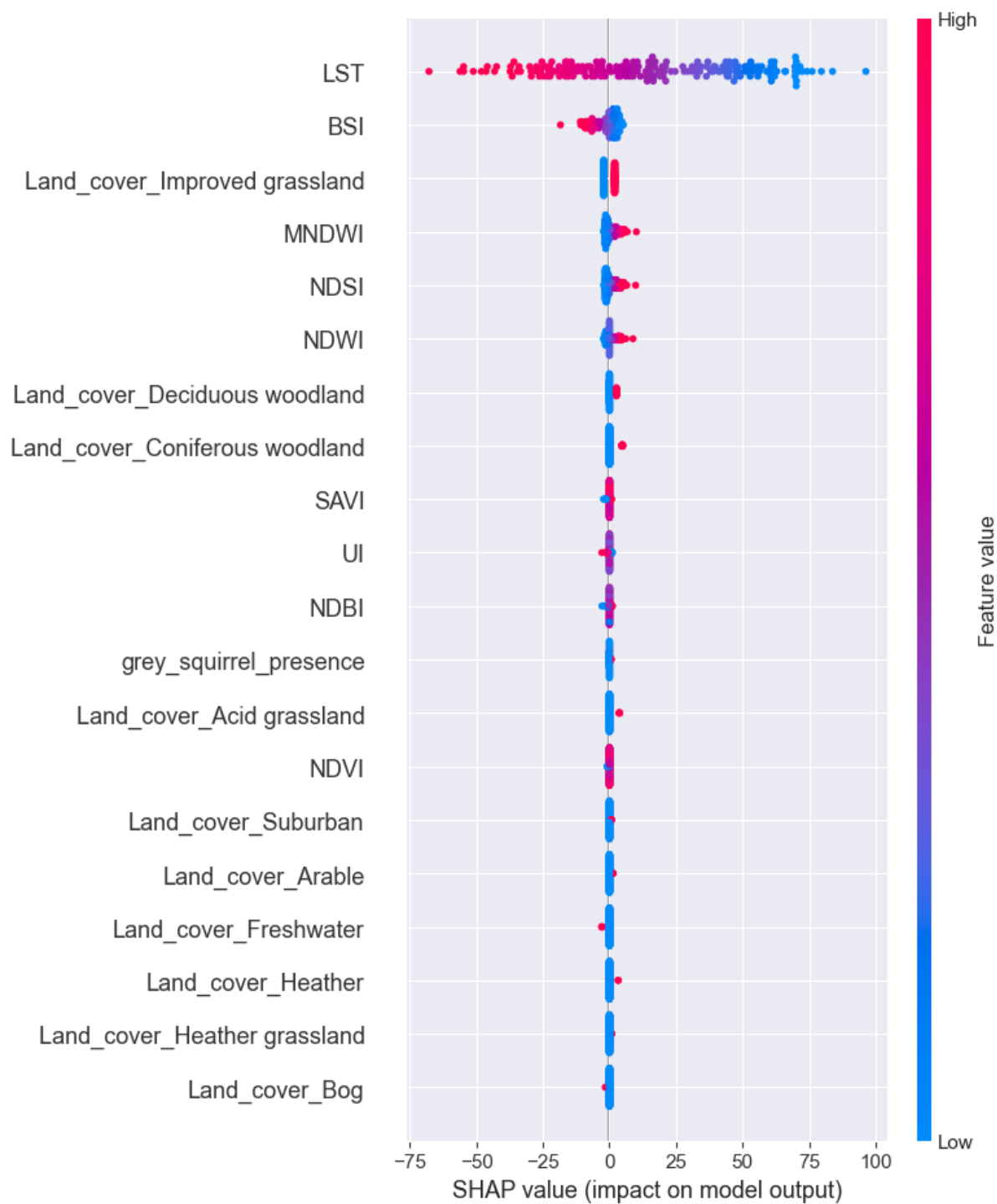


Figure B.9: *Sciurus vulgaris* (Red Squirrel)

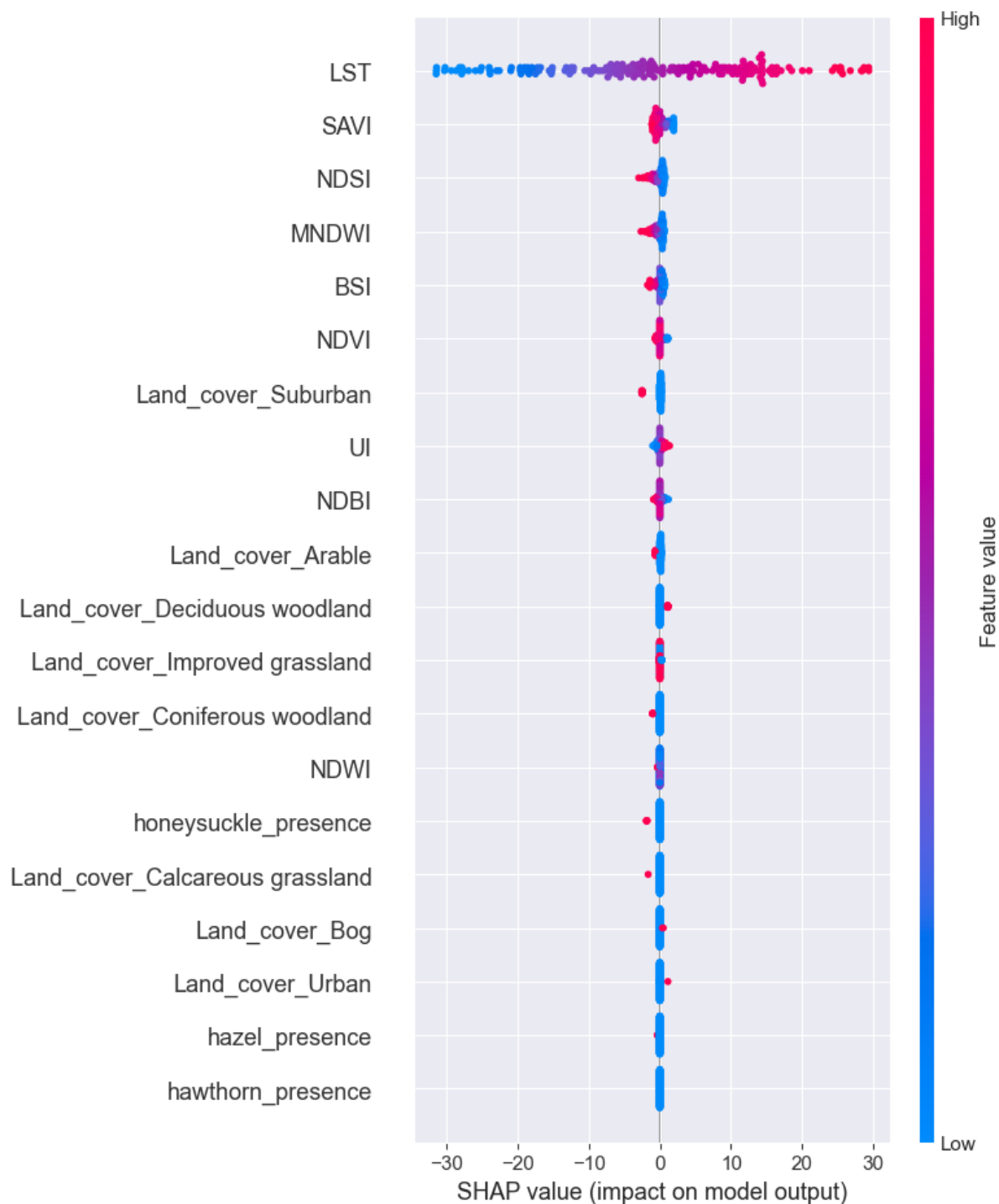


Figure B.10: *Muscardinus avellanarius* (Dormice)

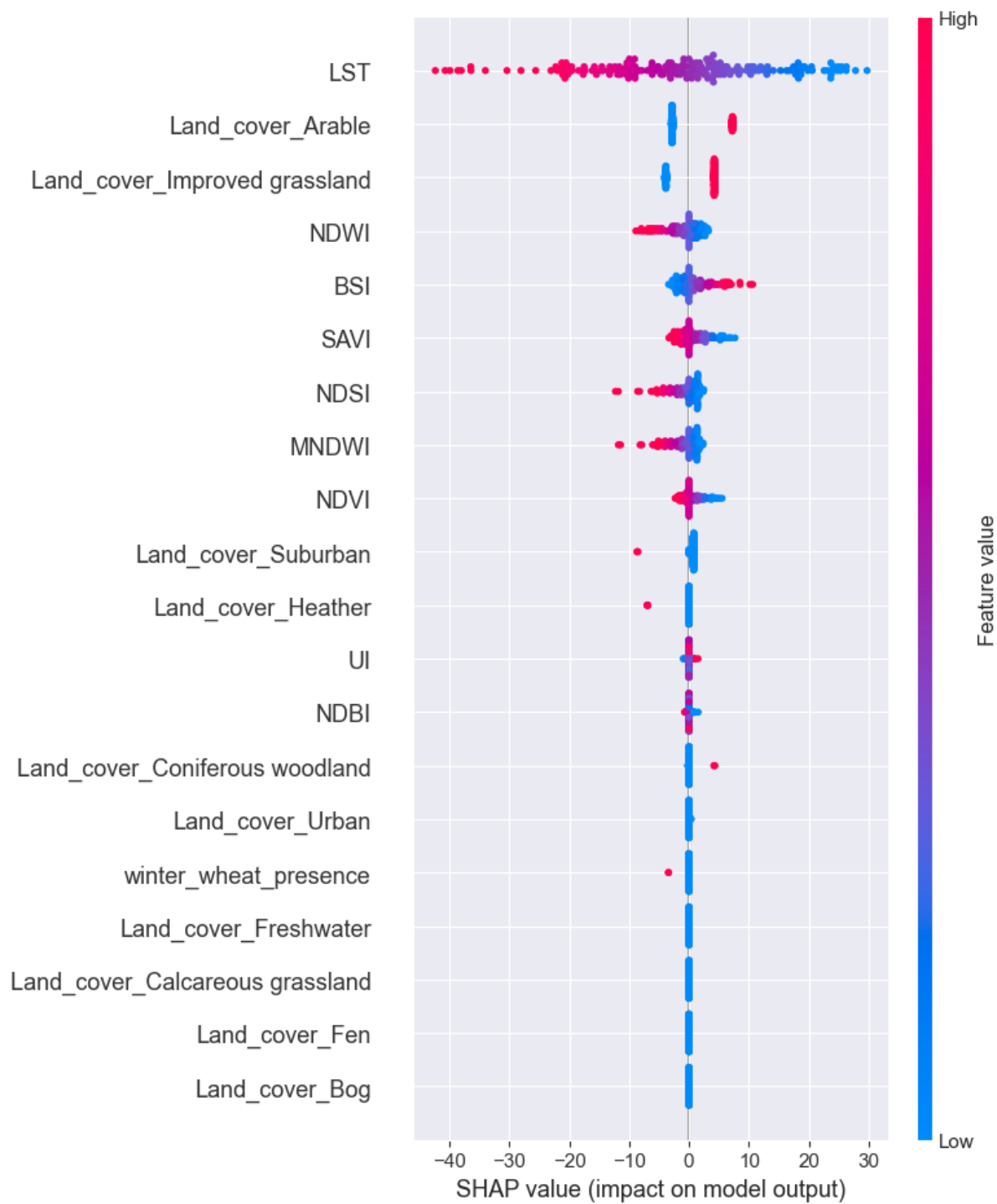


Figure B.11: *Lepus europaeus* (Brown Hare)

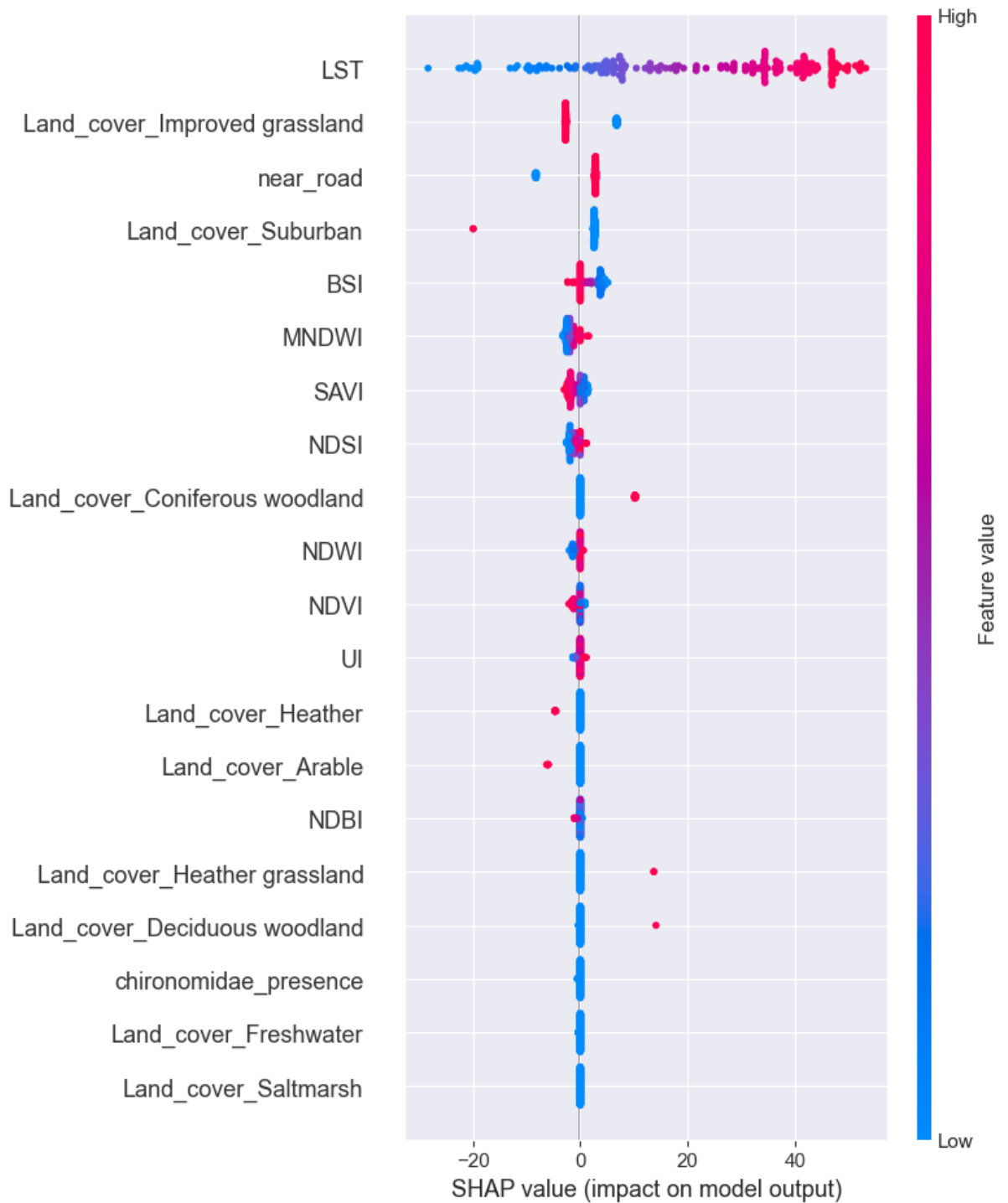


Figure B.12: *Pipistrellus pygmaeus* (Bat)

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